Boca Ciega Bay Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

Contents

Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

With respect to documents and information available from SEACAR DDI, neither the State of Florida nor the Florida Department of Environmental Protection makes any warranty, expressed or implied, including the warranties of merchantability and fitness for a particular purpose arising out of the use or inability to use the data, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights.

This report was funded in part, through a grant agreement from the Florida Department of Environmental Protection, Florida Coastal Management Program, by a grant provided by the Office for Coastal Management under the Coastal Zone Management Act of 1972, as amended, National Oceanic and Atmospheric Administration. The views, statements, findings, conclusions and recommendations expressed herein are those of the author(s) and do not necessarily reflect the views of the State of Florida, NOAA or any of their sub agencies.

Published: 2026-06-10

Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

Parameter Name	*Units*	*Low Threshold*	*High Threshold*
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m^-1	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

SEACAR QAQC Description	*Include*	*SEACAR QAQCFlagCode*
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

Qualifier Source	*Value Qualifier*	*Include*	*MDL*	*Description*
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determiniation; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

Qualifier Source	*Value Qualifier*	*Include*	*Description*
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as El Niño/La Niña, large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p >= 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

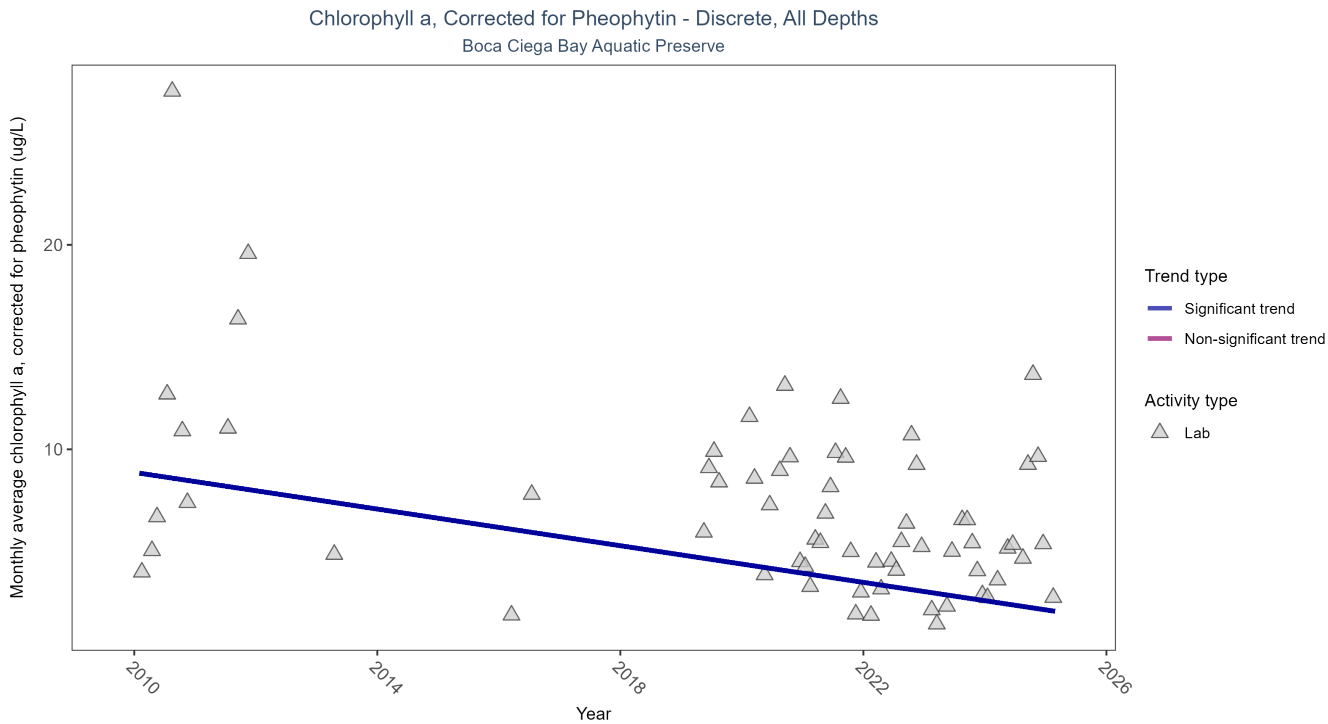

Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	718	11	2010 - 2025	4.6	-0.3556	8.8752	-0.4479	0.0014

Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.45 µg/L per year, indicating an increase in water clarity.

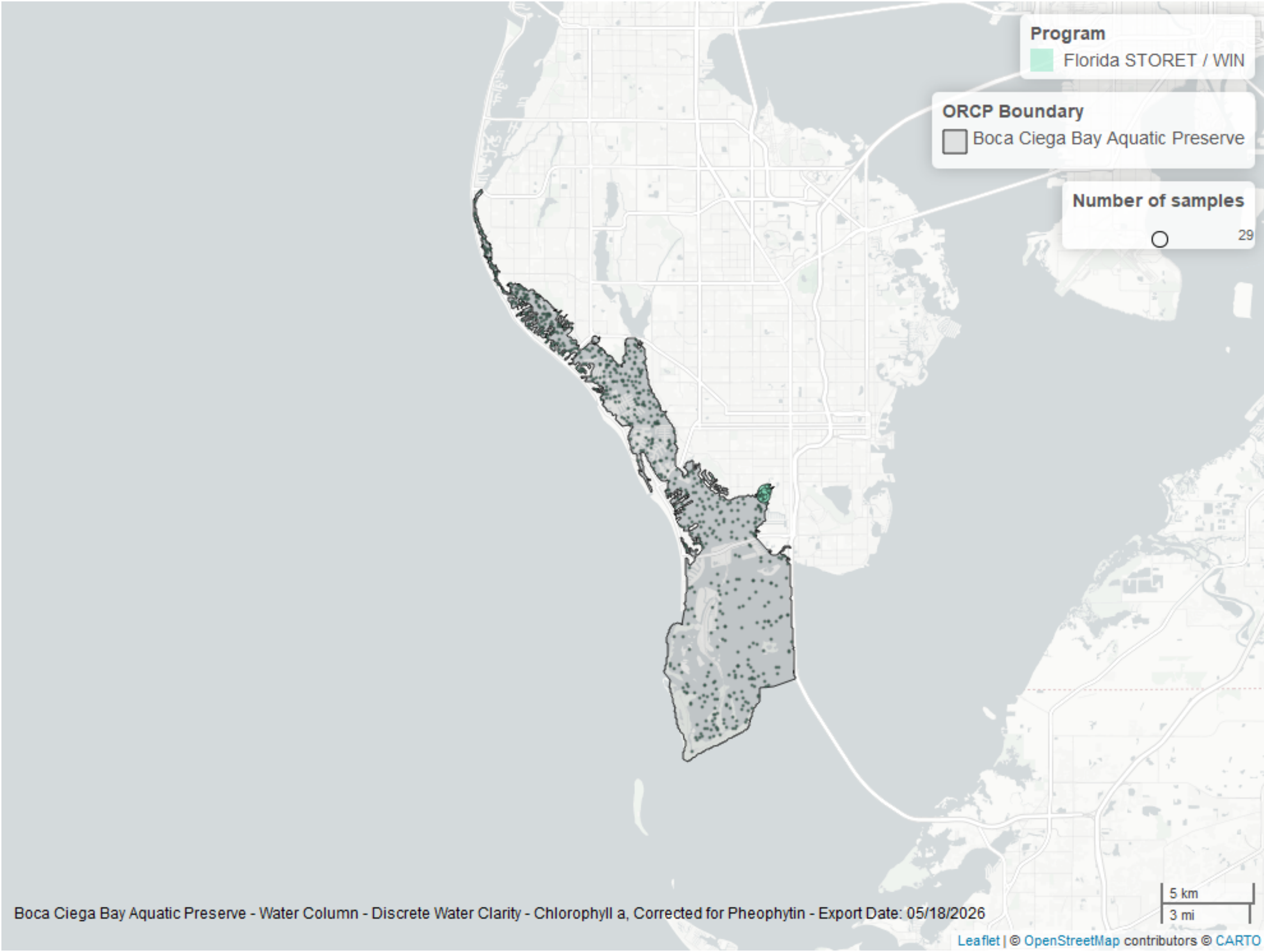

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	733	2010	2025

Program names:

5002 - Florida STORET / WIN[1]

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

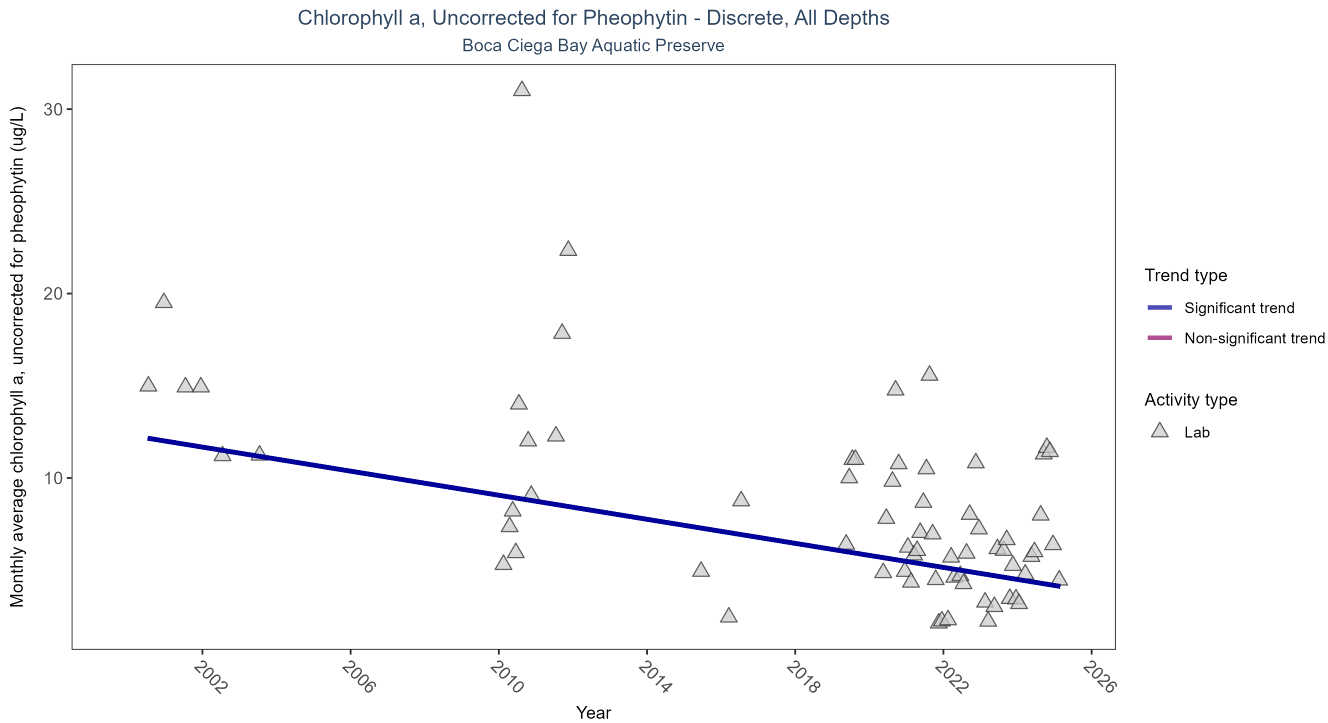

Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	681	15	2000 - 2025	5.4	-0.4047	12.3253	-0.3263	0.0002

Monthly average chlorophyll a, uncorrected for pheophytin, decreased by 0.33 µg/L per year, indicating an increase in water clarity.

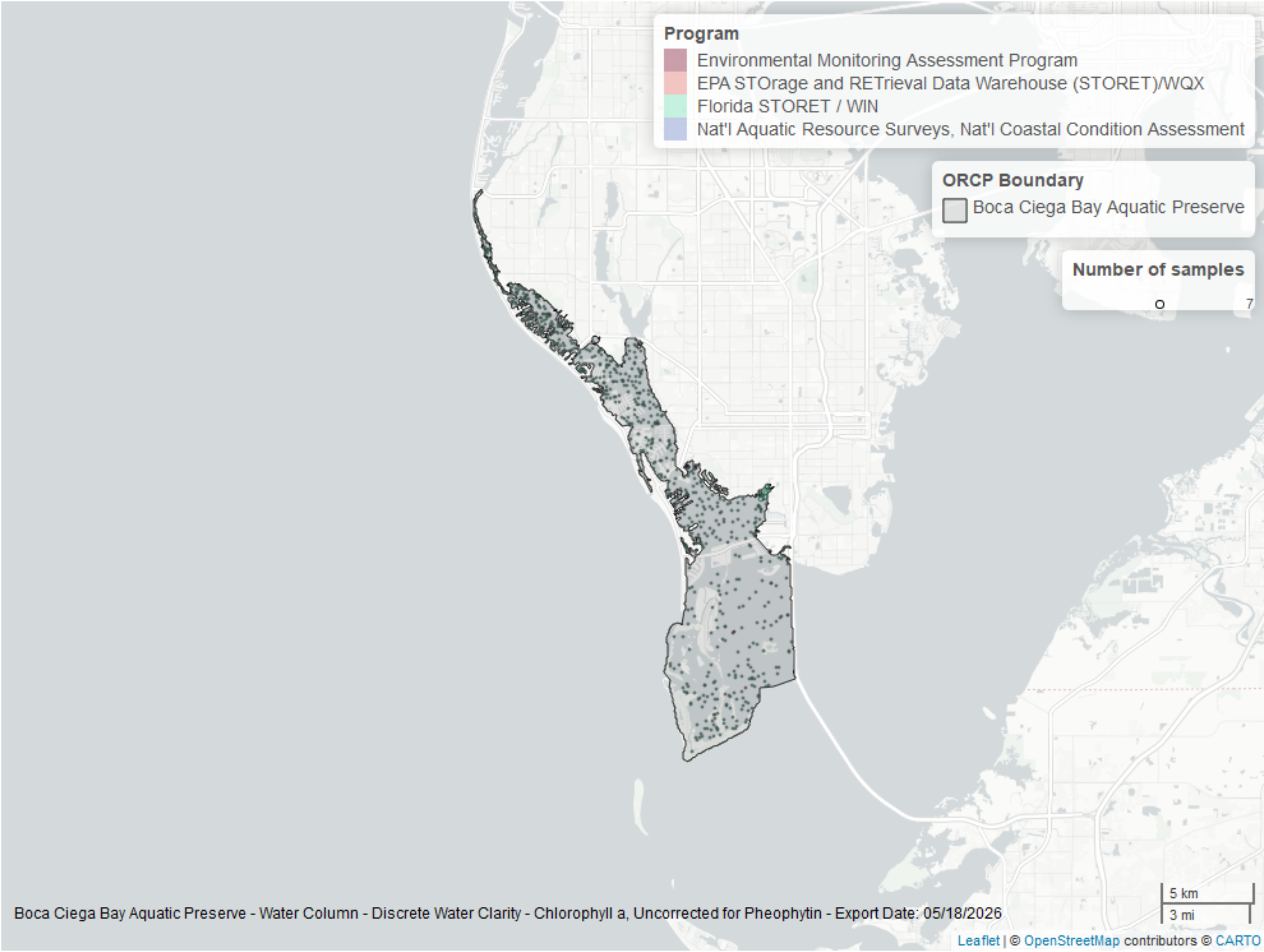

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	667	2010	2025
103	7	2000	2015
115	5	2000	2003
118	3	2000	2010

Program names:

5002 - Florida STORET / WIN[1]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]
115 - Environmental Monitoring Assessment Program[3]
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment[4]

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

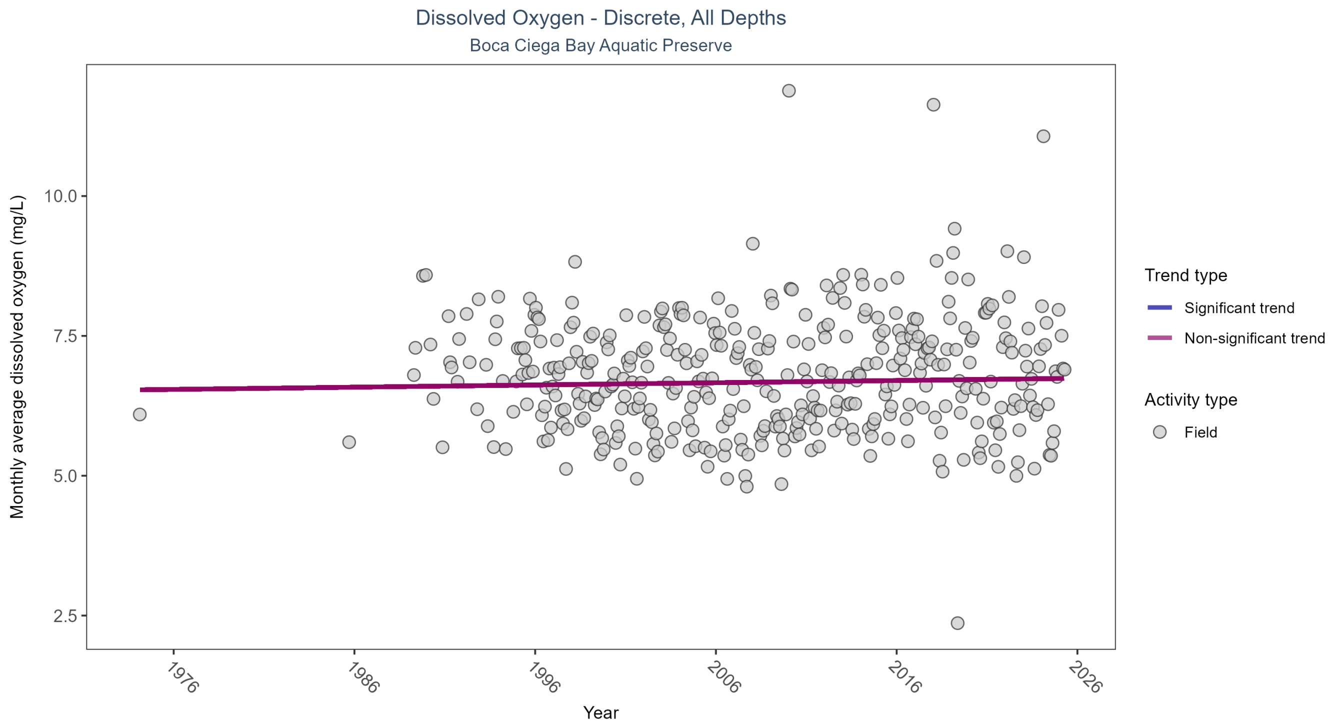

Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	30247	39	1974 - 2025	6.45	0.045	6.5352	0.004	0.2264

Dissolved oxygen showed no detectable trend between 1974 and 2025.

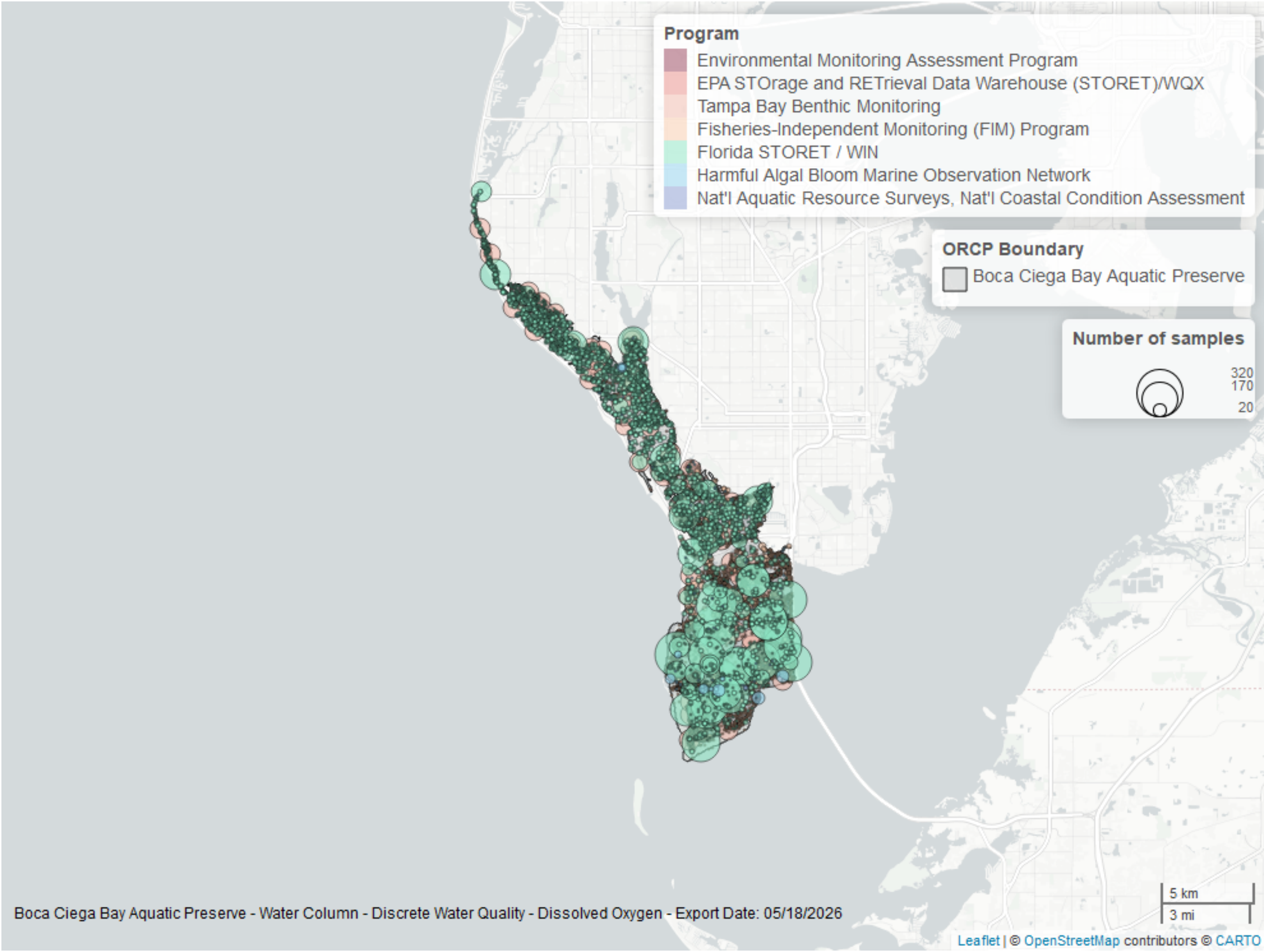

Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	14021	1995	2025
69	10571	1989	2024
4067	5450	1995	2023
95	226	1974	2018
115	10	2000	2003
118	8	2000	2015
103	6	2015	2015

Program names:

5002 - Florida STORET / WIN[1]
69 - Fisheries-Independent Monitoring (FIM) Program[5]
4067 - Tampa Bay Benthic Monitoring[6]
95 - Harmful Algal Bloom Marine Observation Network[7]
115 - Environmental Monitoring Assessment Program[3]

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment[4]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

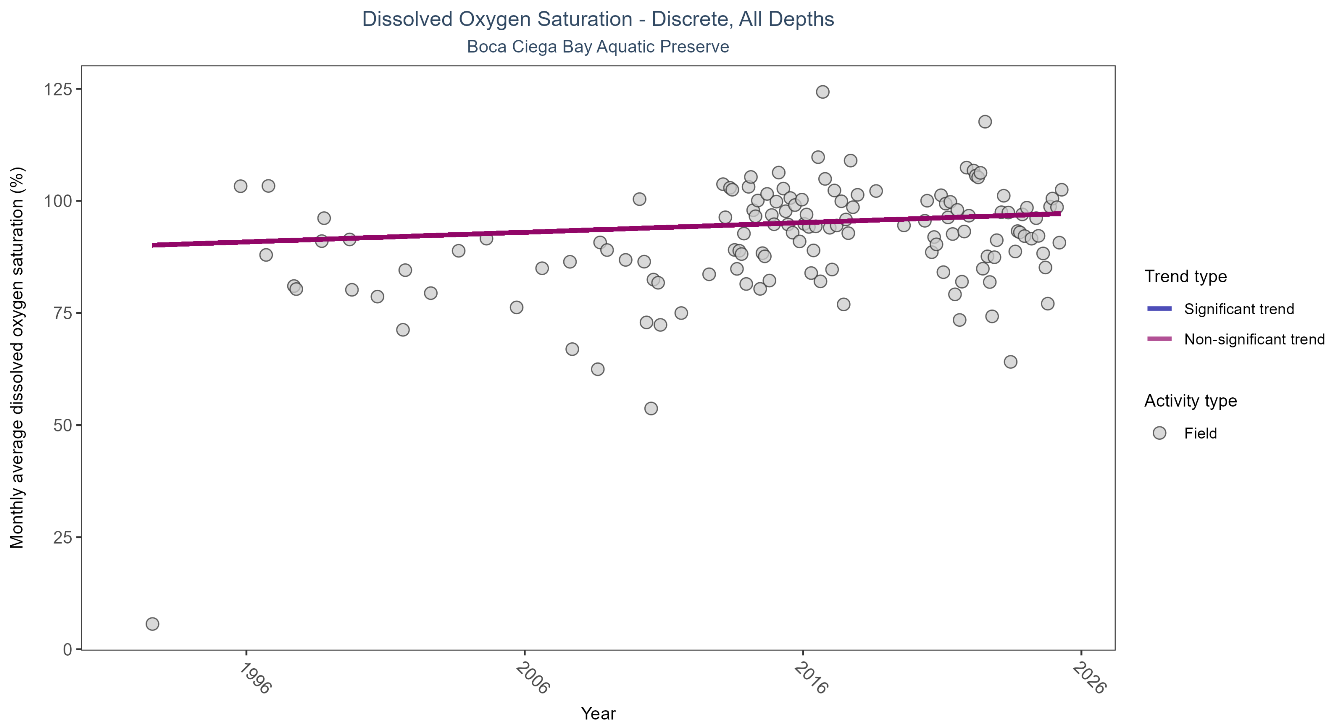

Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	8000	32	1992 - 2025	91.2	0.0872	89.9959	0.2153	0.0641

Dissolved oxygen saturation showed no detectable trend between 1992 and 2025.

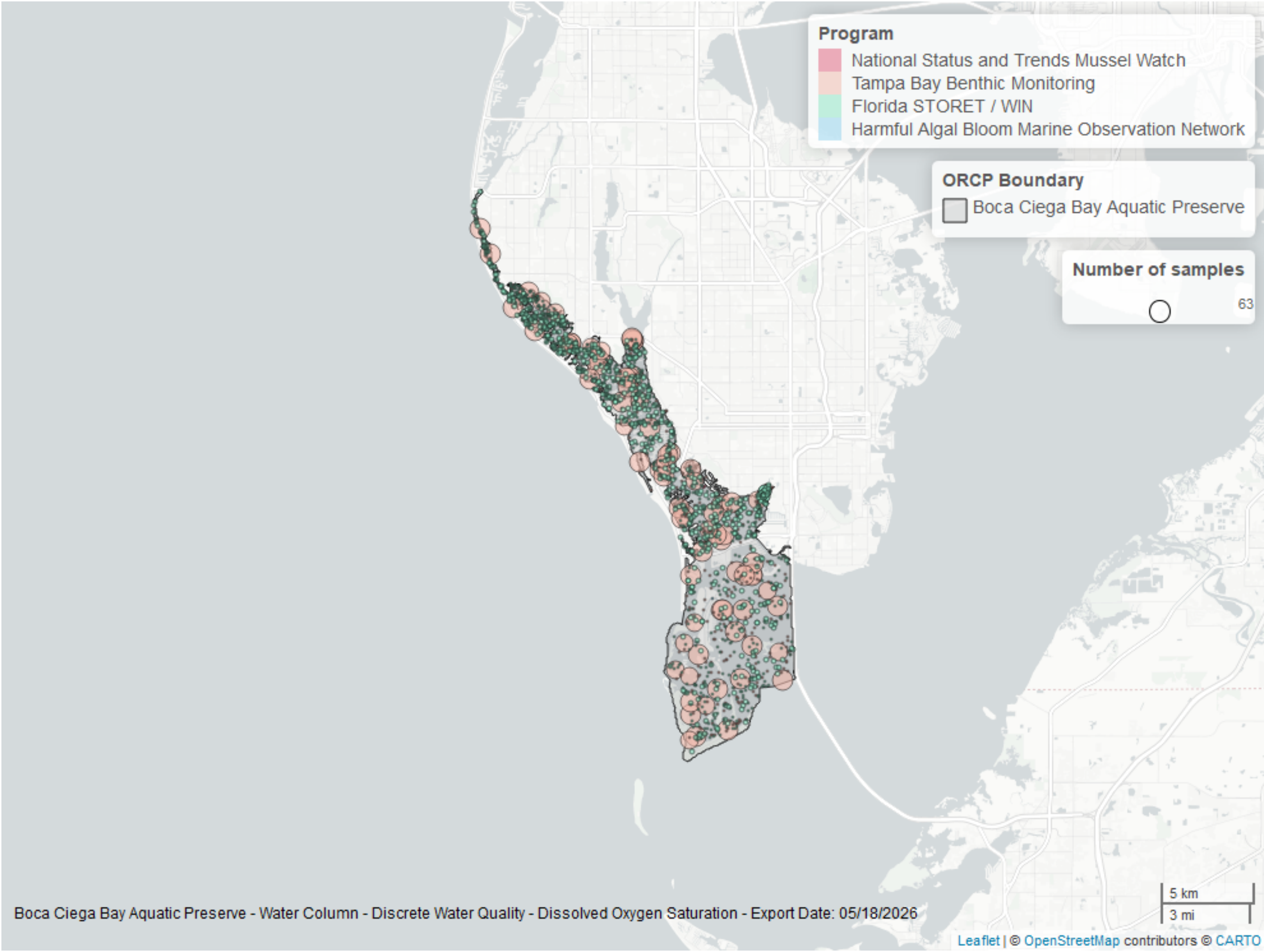

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
4067	4988	1995	2023
5002	3005	2010	2025
102	10	1992	1992
95	1	2017	2017

Program names:

4067 - Tampa Bay Benthic Monitoring[6]
5002 - Florida STORET / WIN[1]
102 - National Status and Trends Mussel Watch[8]
95 - Harmful Algal Bloom Marine Observation Network[7]

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

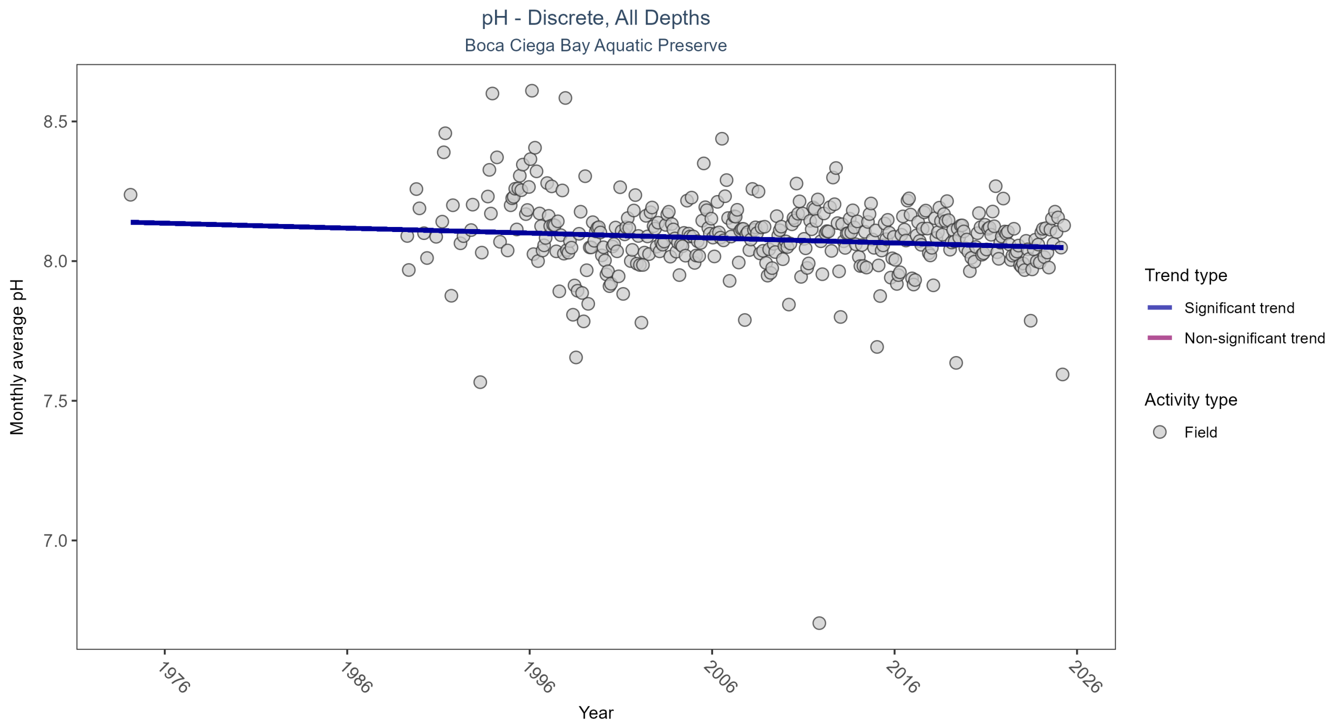

Figure 9: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	27567	38	1974 - 2025	8.1	-0.1144	8.1393	-0.0018	0.0012

Monthly average pH decreased by less than 0.01 pH units per year.

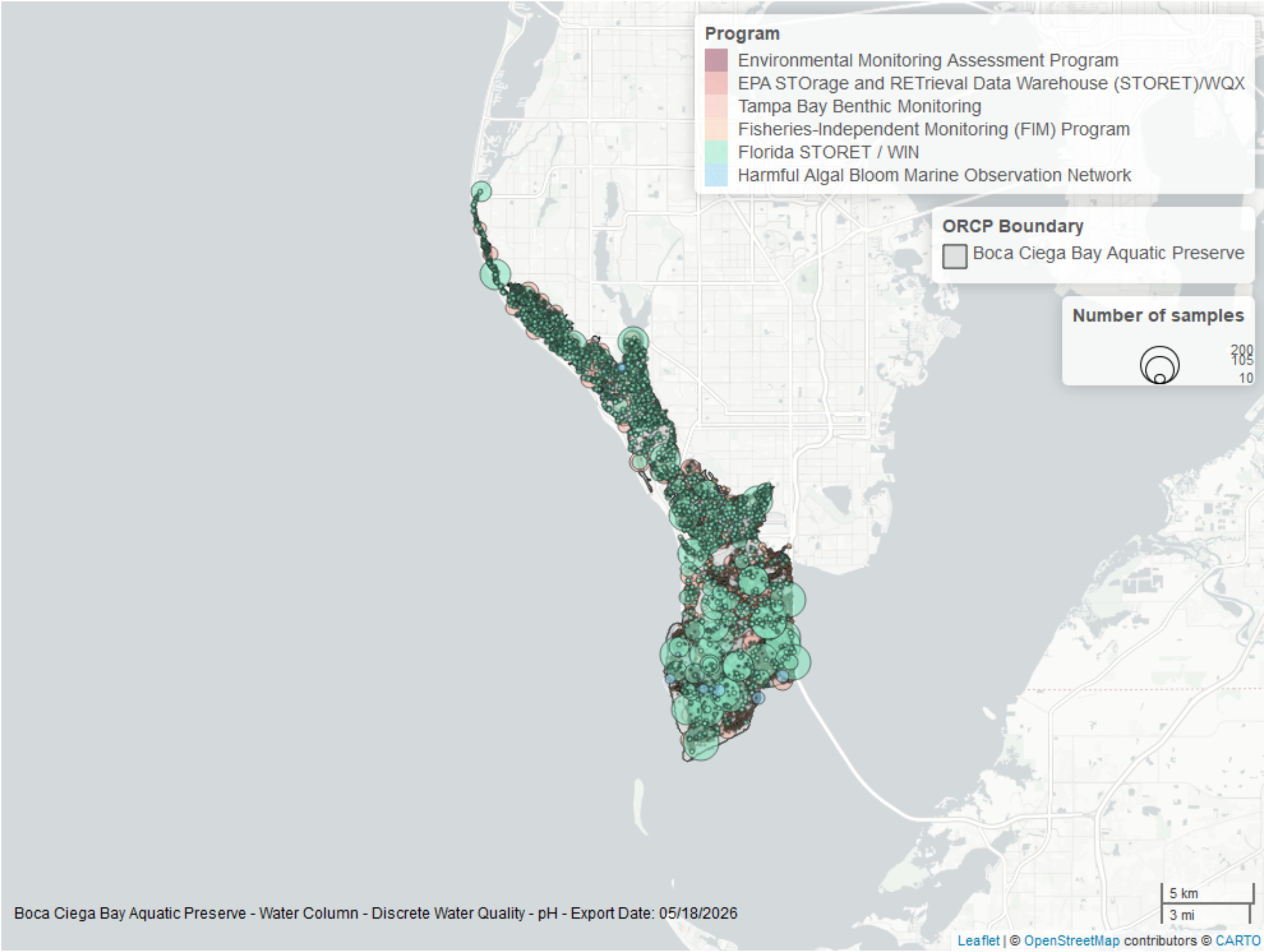

Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	12889	1995	2025
69	10360	1989	2024
4067	4166	1995	2023
95	203	1974	2018
115	10	2000	2003
103	3	2015	2015

Program names:

5002 - Florida STORET / WIN[1]
69 - Fisheries-Independent Monitoring (FIM) Program[5]
4067 - Tampa Bay Benthic Monitoring[6]
95 - Harmful Algal Bloom Marine Observation Network[7]
115 - Environmental Monitoring Assessment Program[3]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

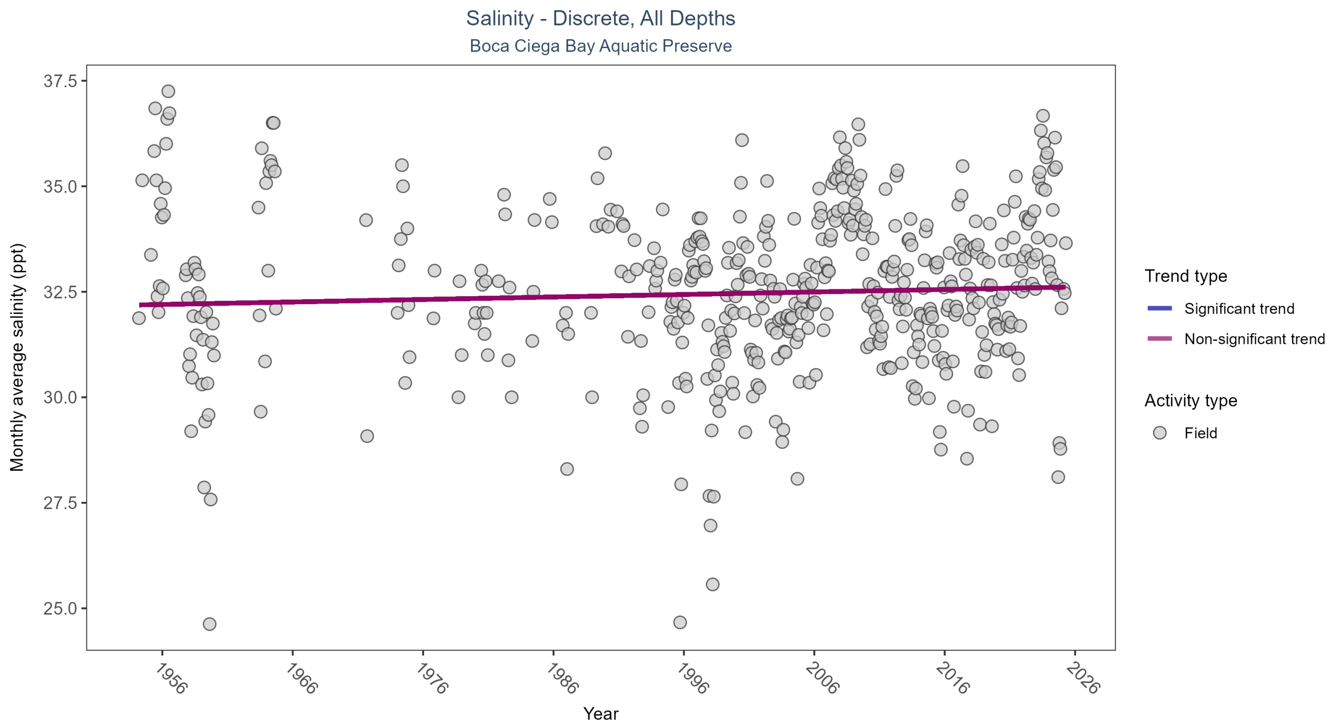

Figure 11: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	27920	58	1954 - 2025	32.67	0.0457	32.1871	0.0059	0.1735

Salinity showed no detectable trend between 1954 and 2025.

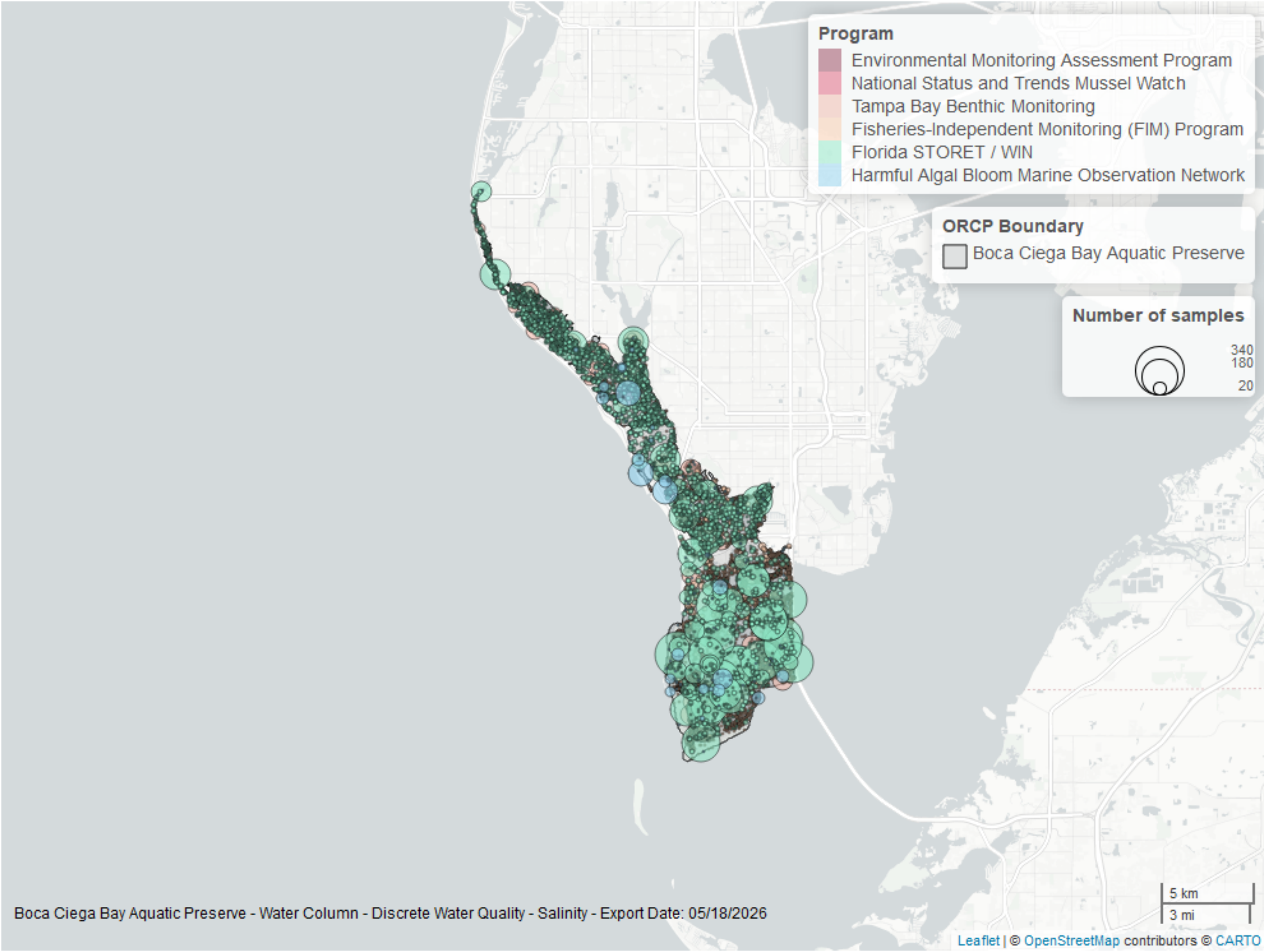

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	14044	1995	2025
69	10602	1989	2024
4067	2498	1995	2023
95	805	1954	2018
115	10	2000	2003
102	10	1992	1992

Program names:

5002 - Florida STORET / WIN[1]
69 - Fisheries-Independent Monitoring (FIM) Program[5]
4067 - Tampa Bay Benthic Monitoring[6]
95 - Harmful Algal Bloom Marine Observation Network[7]
115 - Environmental Monitoring Assessment Program[3]
102 - National Status and Trends Mussel Watch[8]

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

1) TN = TKN + NO3O2;
2) TN = TKN + NO3 + NO2;
3) TN = ORGN + NH4 + NO3O2;
4) TN = ORGN + NH4 + NO2 + NO3;
5) TN = TKN + NO3;
6) TN = ORGN + NH4 + NO3;

Additional Information:

- Rules for use of sample fraction:
  - Florida Department of Environmental Protection (FDEP) report that if both "Total" and "Dissolved" components are reported, only "Total" is used. If the total is not reported, then the dissolved components are used as a best available replacement.
  - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
  - ParameterName = "Total Nitrogen"
  - SEACAR_QAQCFlagCode = "1Q"
  - SEACAR_QAQC_Description = "SEACAR Calculated"

Seasonal Kendall-Tau Trend Analysis

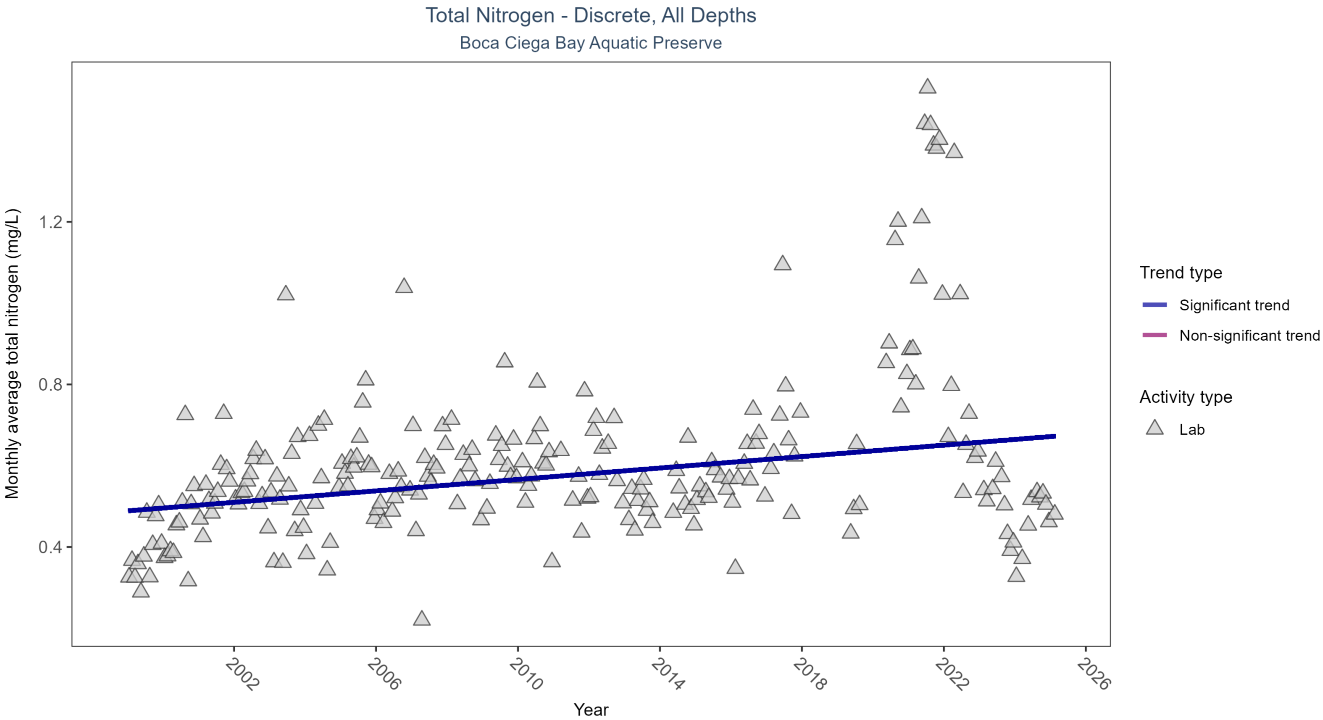

Figure 13: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2977	26	1999 - 2025	0.55	0.2495	0.489	0.007	0

Monthly average total nitrogen increased by 0.01 mg/L per year.

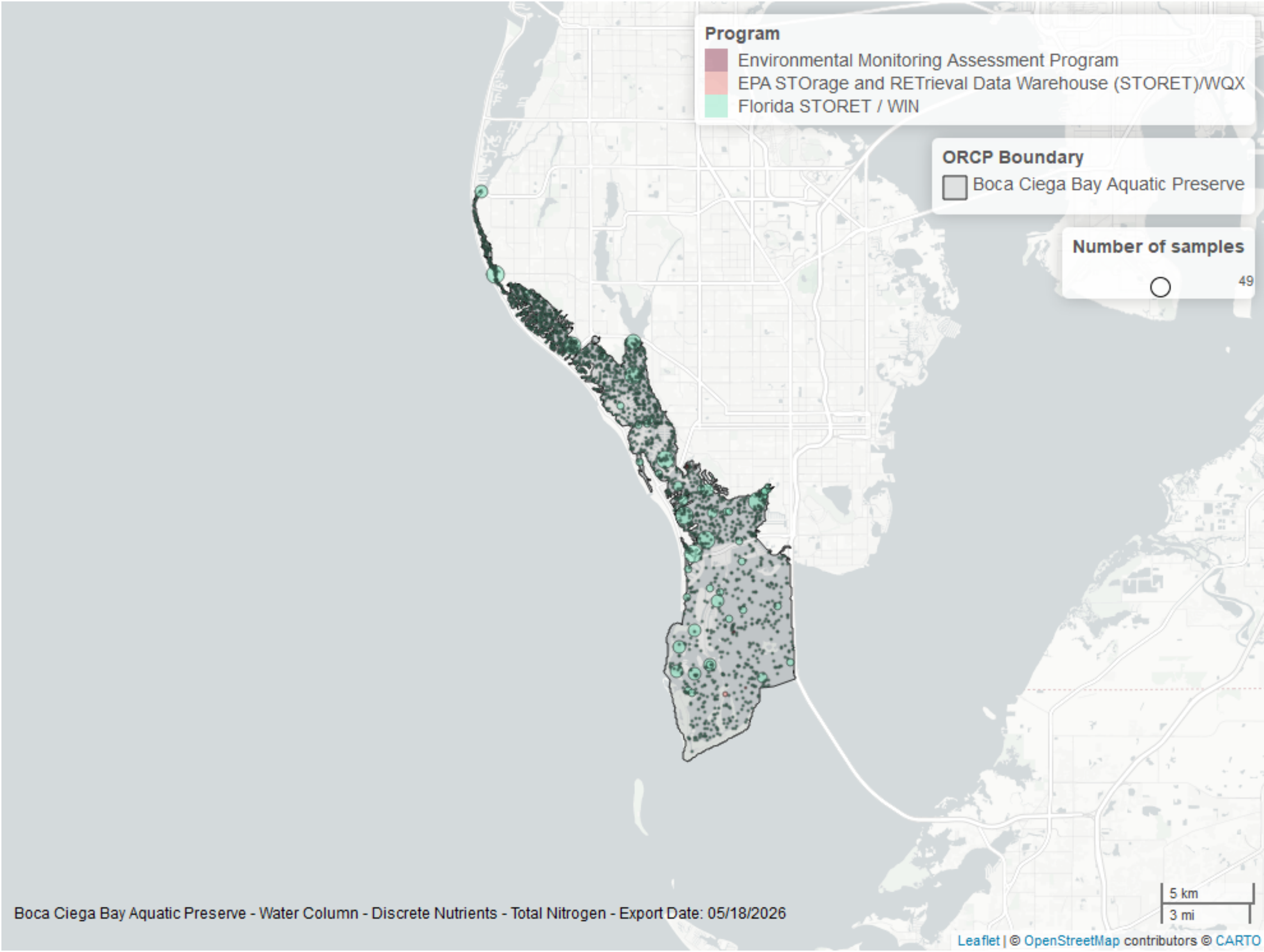

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
5002	2960	1999	2025
103	12	2000	2015
115	5	2000	2003

Program names:

5002 - Florida STORET / WIN[1]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]
115 - Environmental Monitoring Assessment Program[3]

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

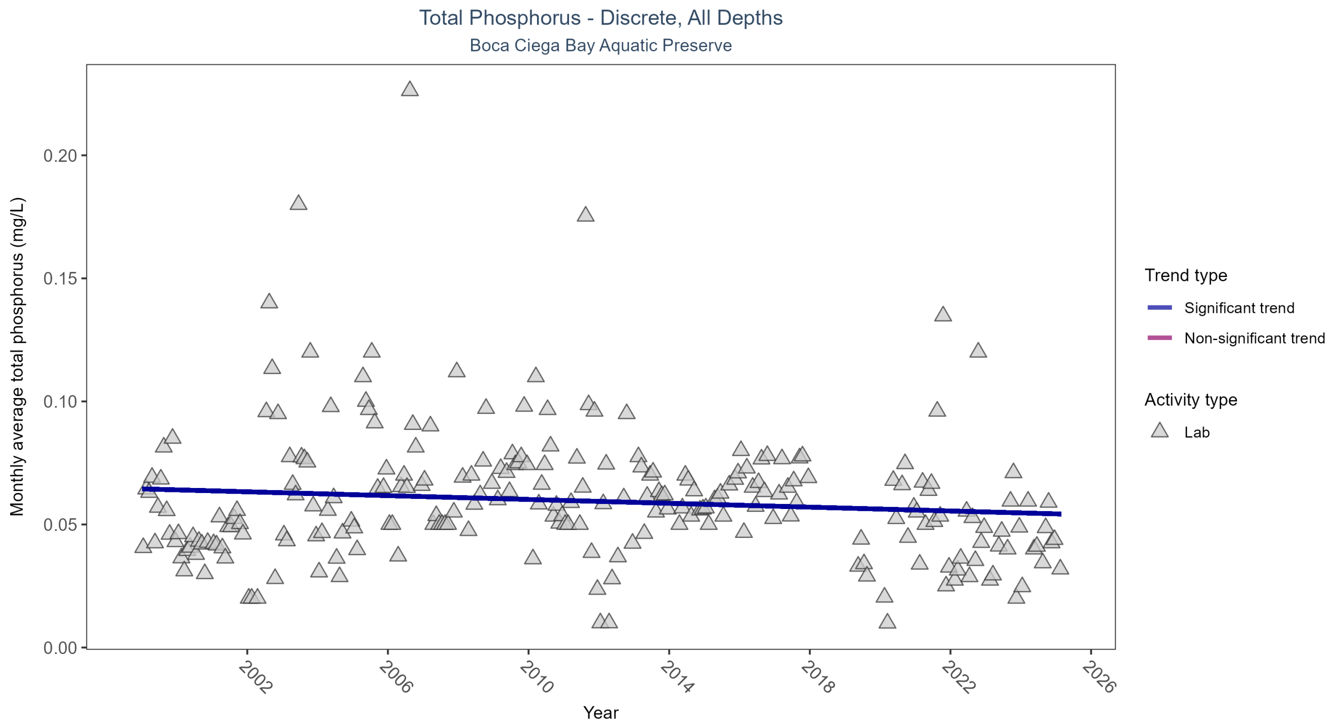

Figure 15: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	2737	26	1999 - 2025	0.05	-0.0986	0.0644	-0.0004	0.0234

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

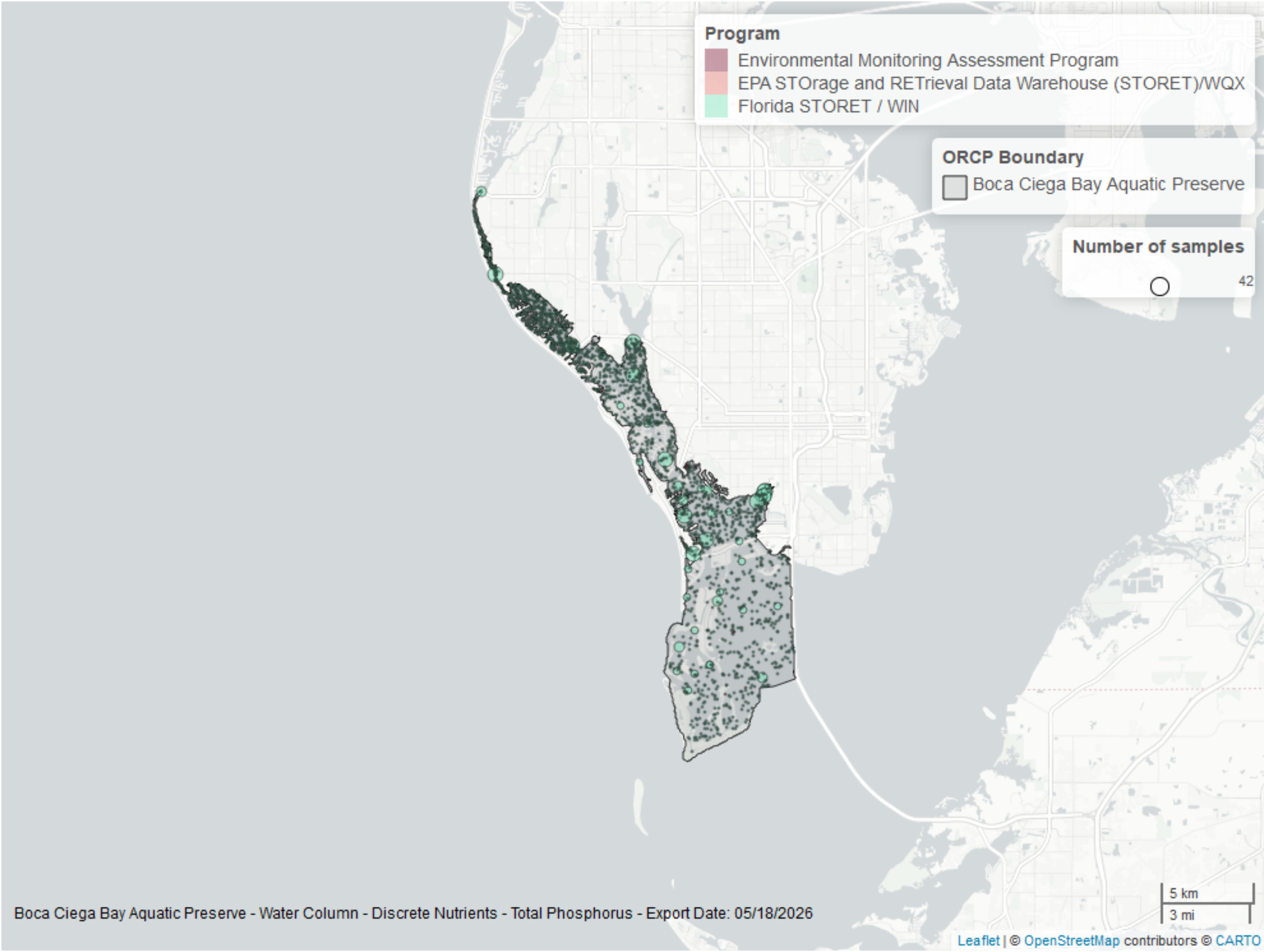

Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5002	2757	1999	2025
103	10	2000	2015
115	5	2000	2003

Program names:

5002 - Florida STORET / WIN[1]
103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX[2]
115 - Environmental Monitoring Assessment Program[3]

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

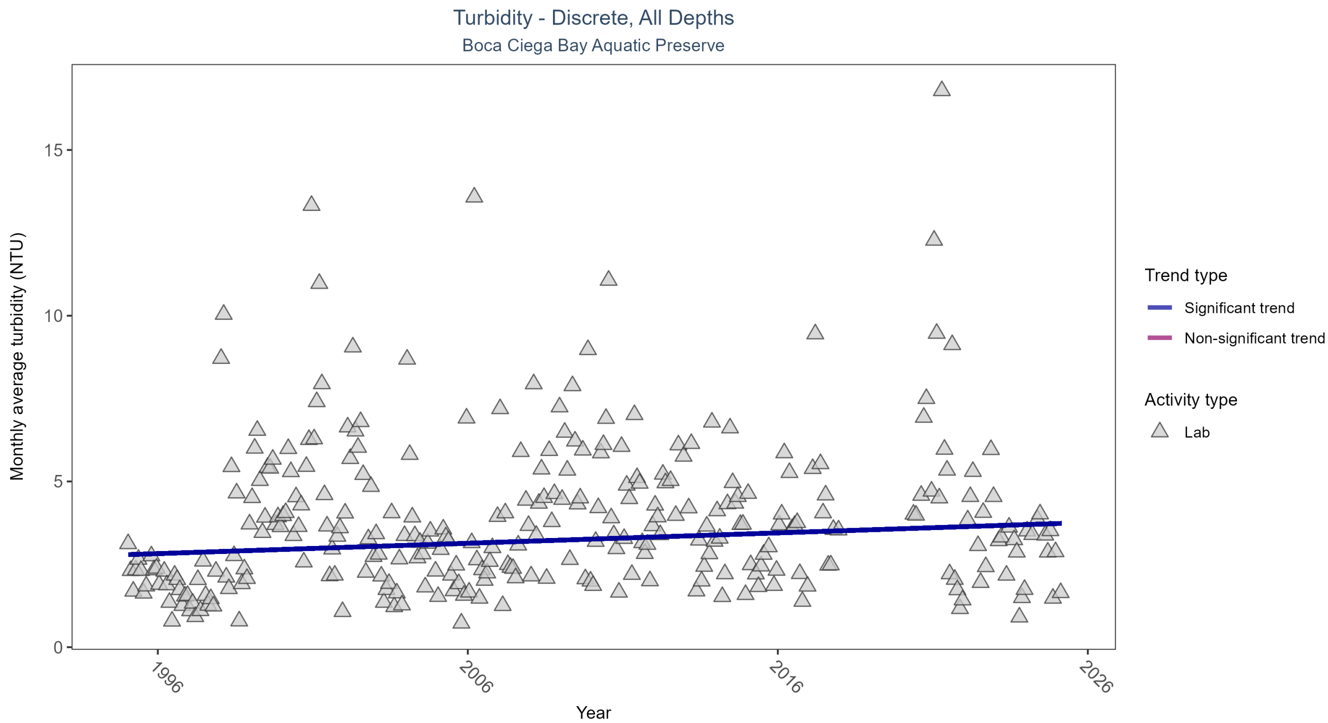

Figure 17: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	7052	29	1995 - 2025	2.6	0.1119	2.7911	0.0313	0.0053

Monthly average turbidity increased by 0.03 NTU per year, indicating a decrease in water clarity.

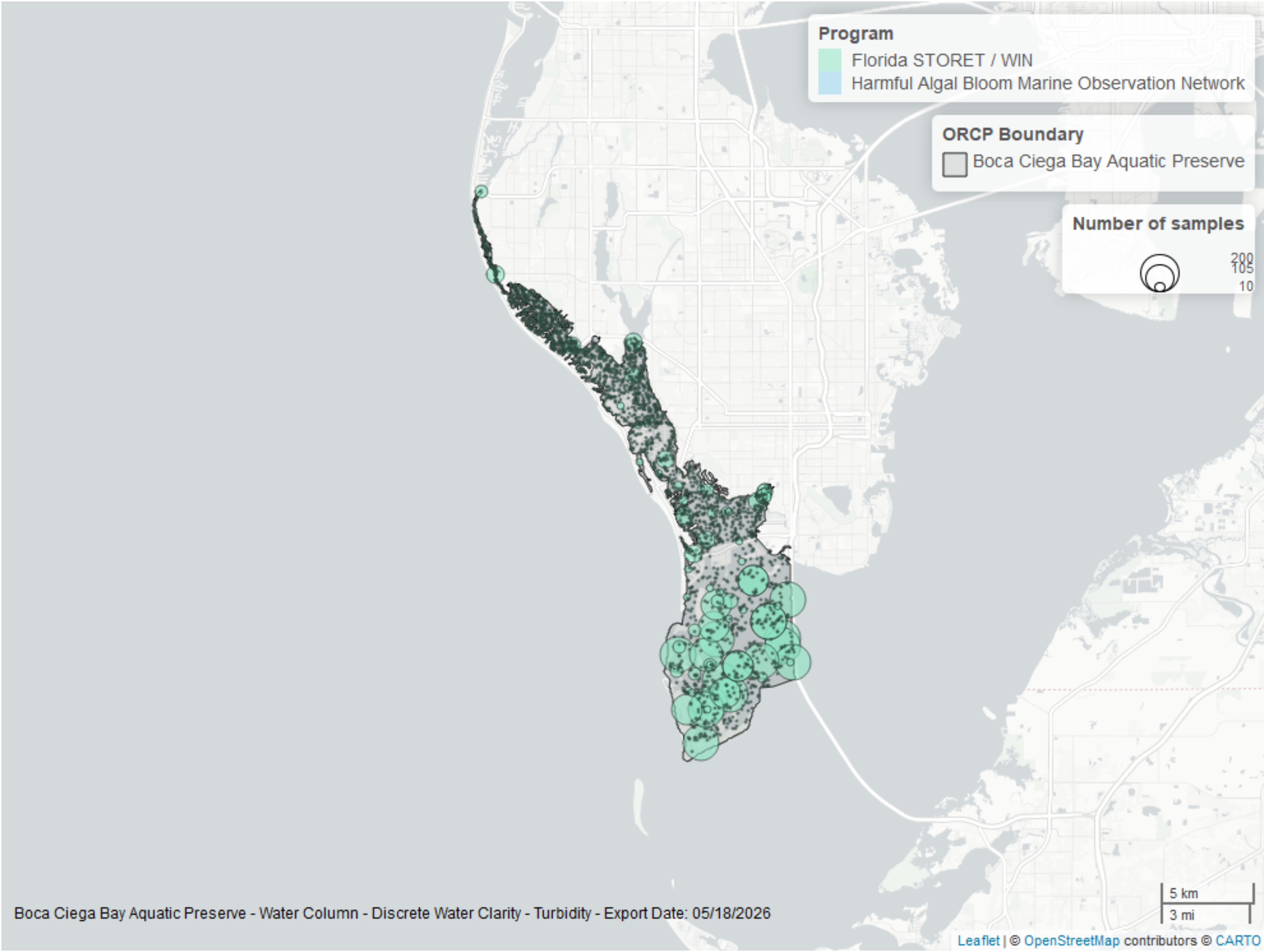

Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	7145	1995	2025
95	2	2004	2004

Program names:

5002 - Florida STORET / WIN[1]
95 - Harmful Algal Bloom Marine Observation Network[7]

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

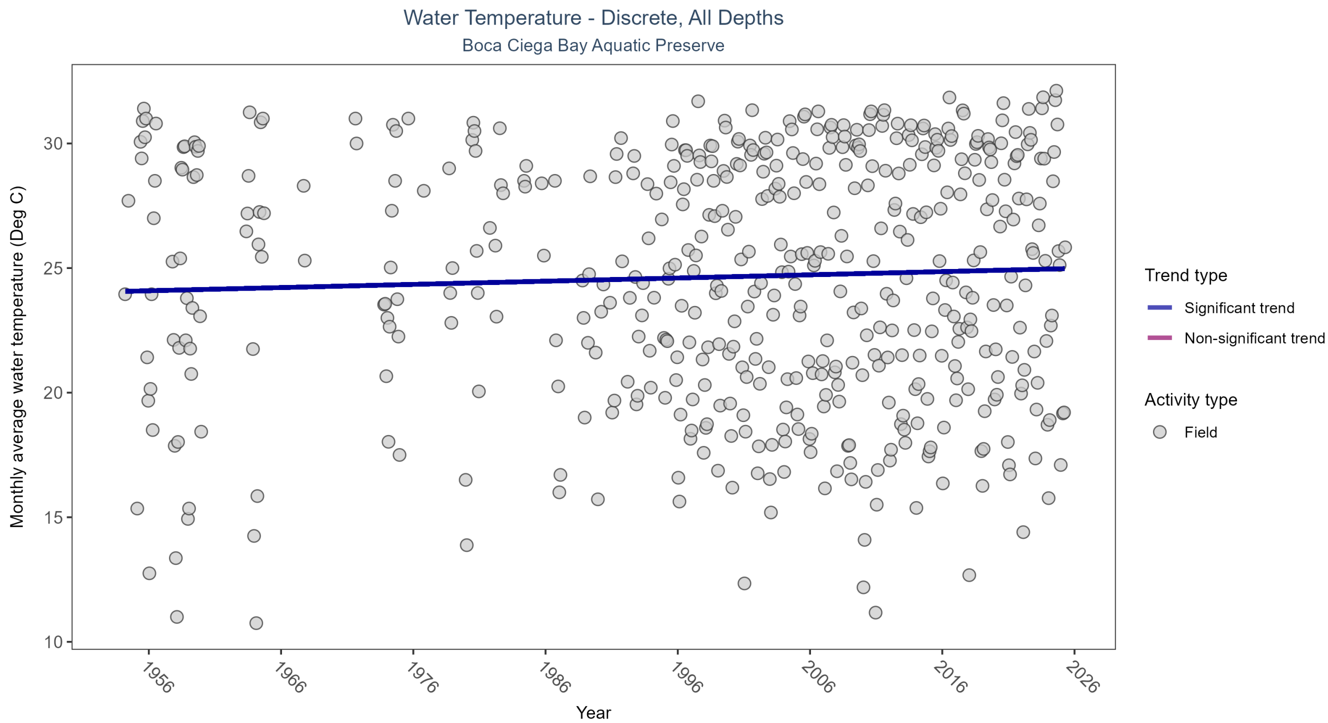

Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	30713	61	1954 - 2025	27.1	0.0994	24.0633	0.0128	0.0015

Monthly average water temperature increased by 0.01°C per year.

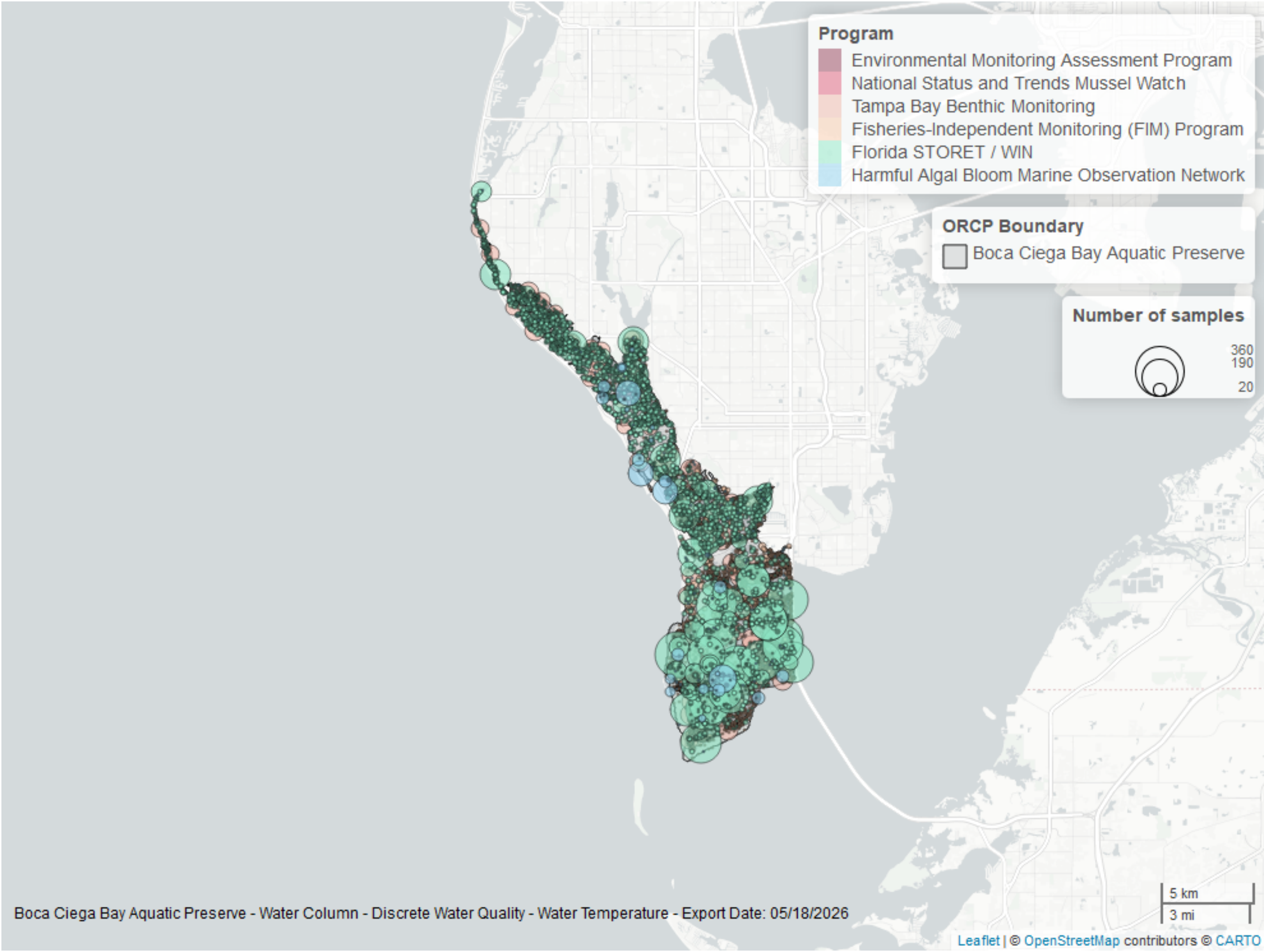

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	14301	1995	2025
69	10624	1989	2024
4067	4904	1995	2023
95	865	1954	2018
115	10	2000	2003
102	10	1992	1992

Program names:

5002 - Florida STORET / WIN[1]
69 - Fisheries-Independent Monitoring (FIM) Program[5]
4067 - Tampa Bay Benthic Monitoring[6]
95 - Harmful Algal Bloom Marine Observation Network[7]
115 - Environmental Monitoring Assessment Program[3]
102 - National Status and Trends Mussel Watch[8]

Submerged Aquatic Vegetation

The data file used is: **All__SAV__Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (*SAV*) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of *SAV* inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover."

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila spp.* are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

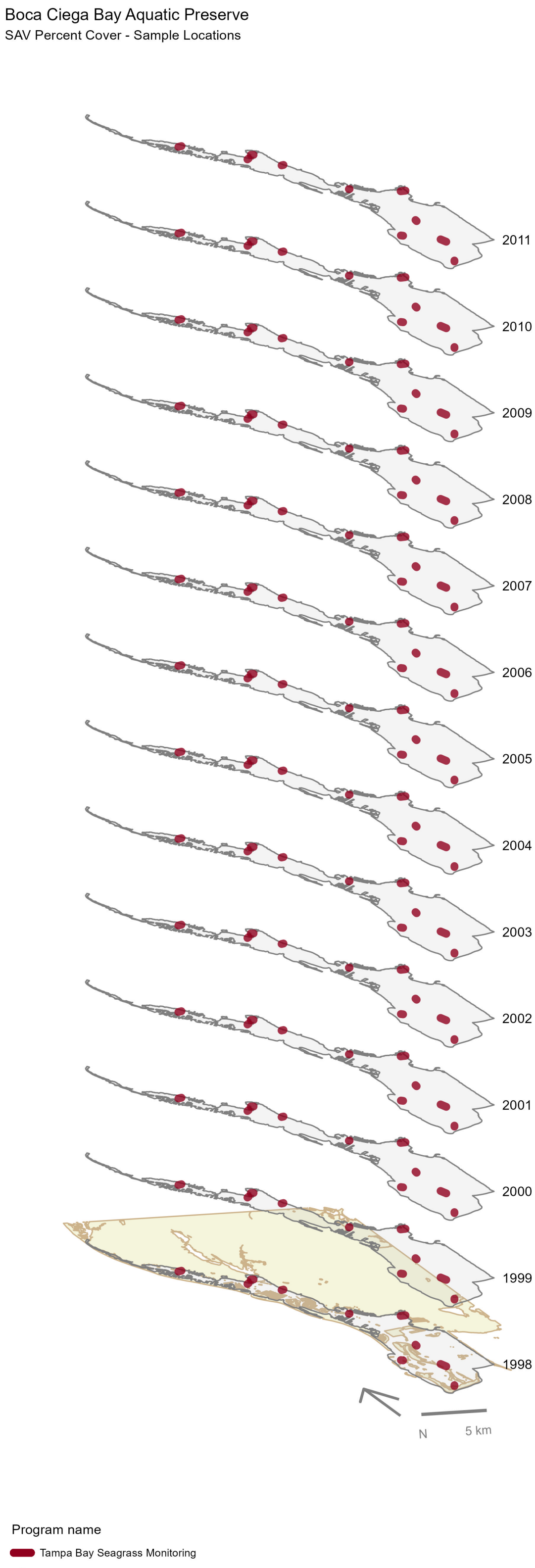

Figure 21: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Boca Ciega Bay Aquatic Preserve* by program name.

Click here to view spatio-temporal plots on GitHub.

Sampling locations by Program:

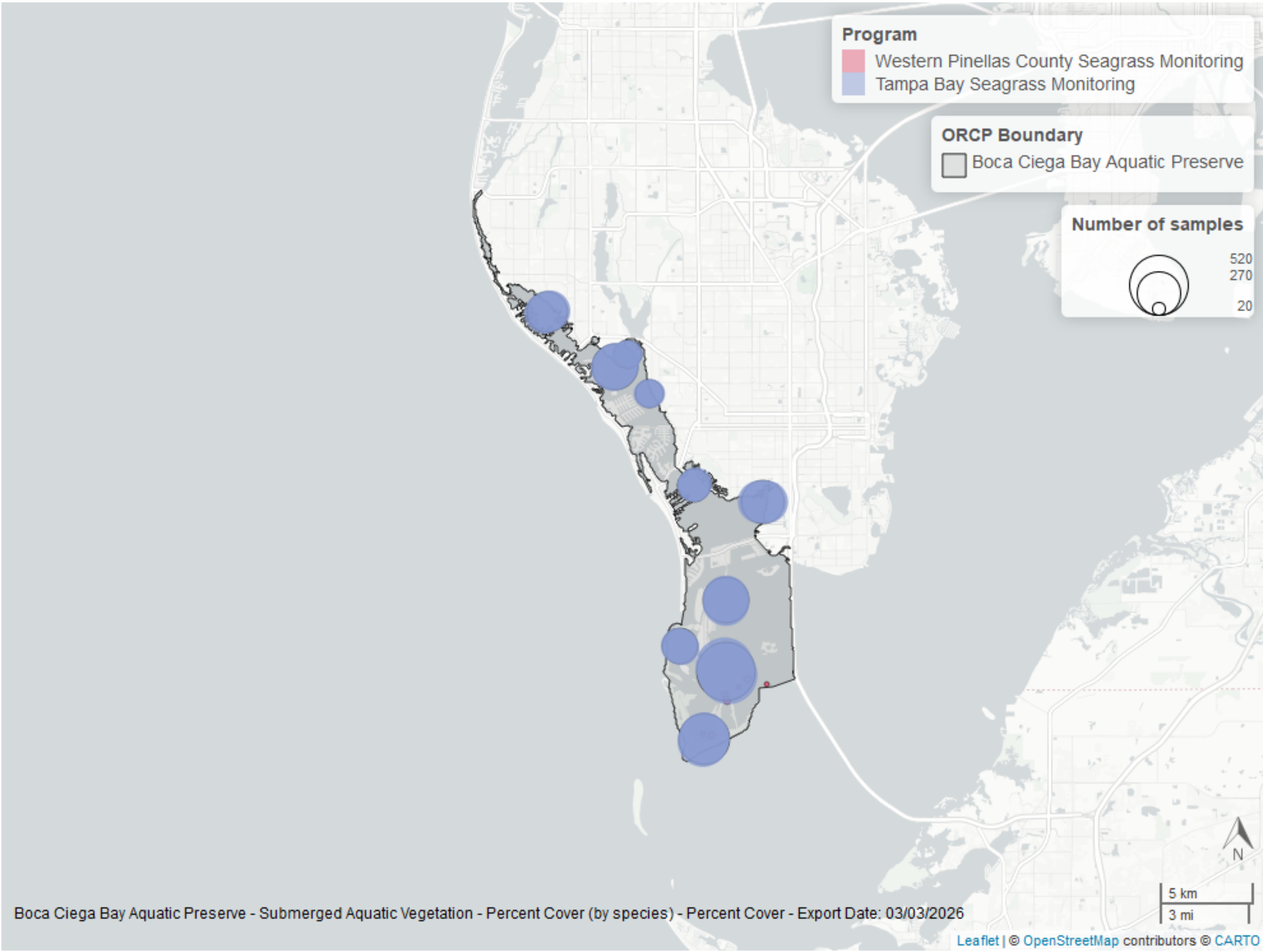

Figure 22: Map showing SAV sampling sites within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 26: Program Information for Submerged Aquatic Vegetation

ProgramID	*N-Data*	*YearMin*	*YearMax*	*method*	*Sample Locations*
565	2730	1998	2024	Braun Blanquet	10
564	54	2019	2019	Percent Cover	7

Program names:

564 - Western Pinellas County Seagrass Monitoring[9]
565 - Tampa Bay Seagrass Monitoring[10]

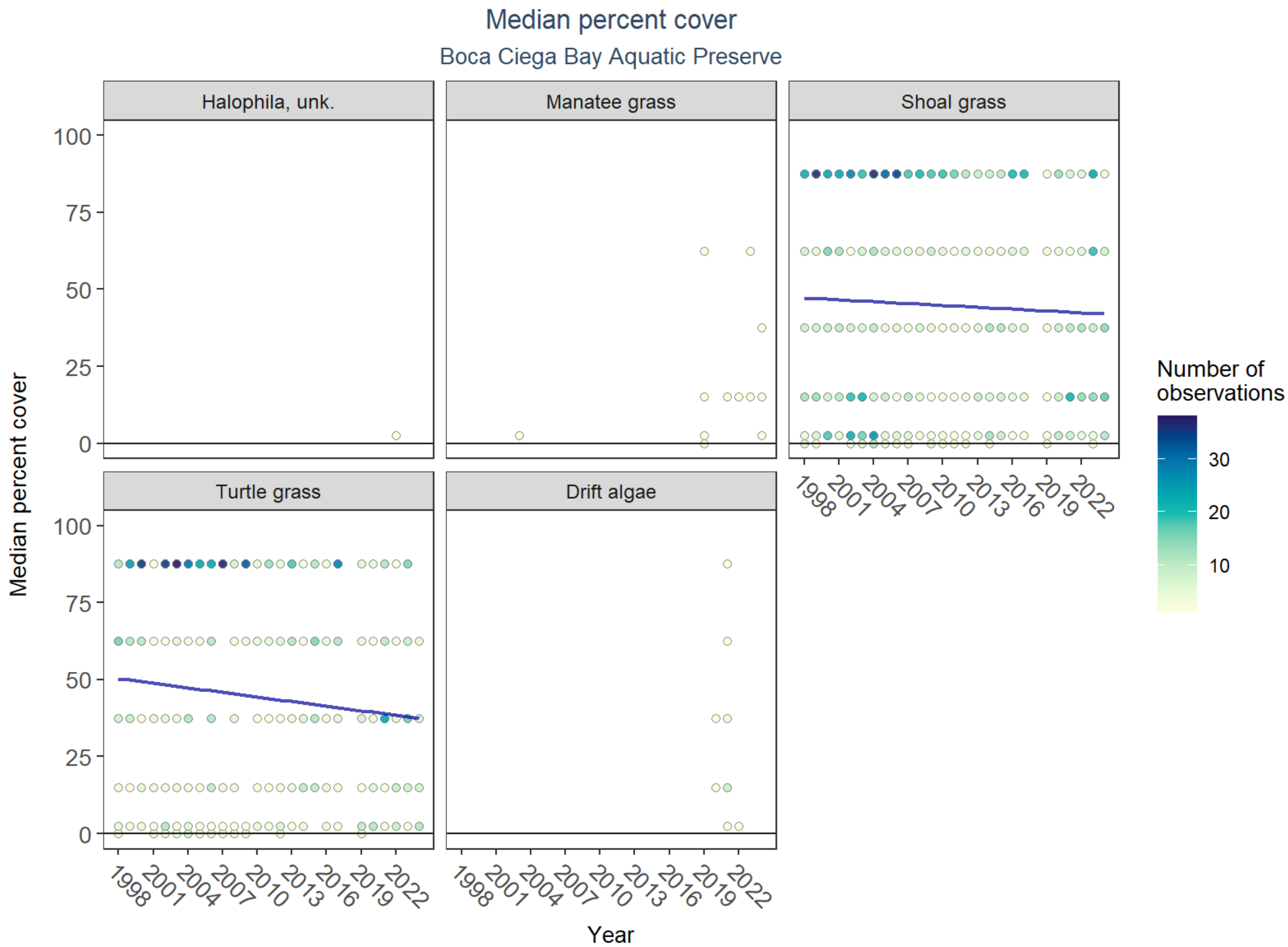

Figure 23: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

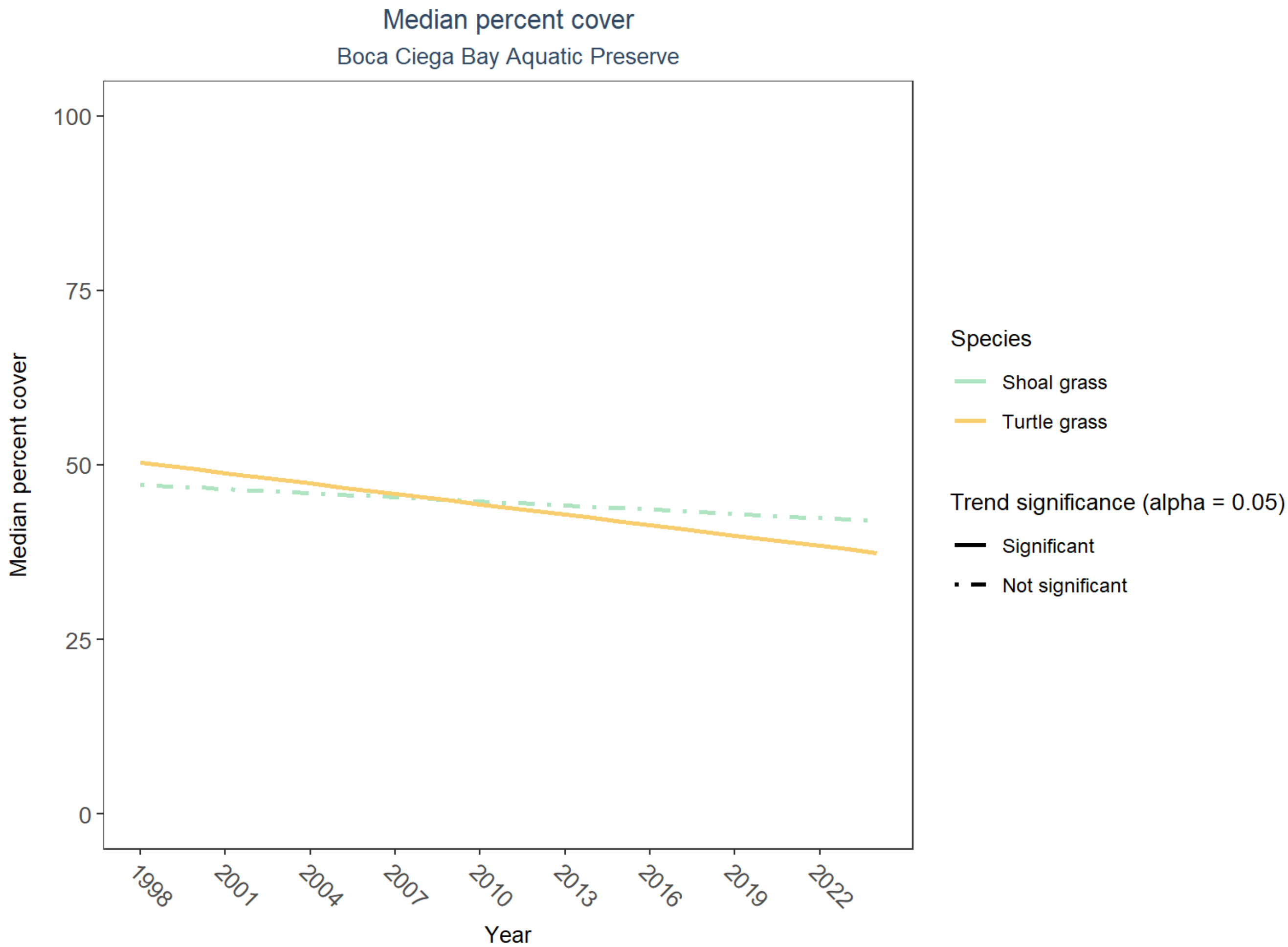

Figure 24: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 27: Percent Cover Trend Analysis for Boca Ciega Bay Aquatic Preserve

Species	*Statistical Trend*	*Period of Record*	*LME Intercept*	*LME Slope*	*P*
Drift algae	Insufficient data	-	-	-	-
Shoal grass	No detectable trend	1998 - 2024	47.97762	-0.1982890	0.3776091
Manatee grass	Model did not fit the available data	2003 - 2024	-	-	-
Turtle grass	Decreasing trend	1998 - 2024	52.32672	-0.4964616	0.0491915
Halophila, unk.	Insufficient data	-	-	-	-

An annual decrease in percent cover was observed for turtle grass (-0.50%). No detectable change in percent cover was observed for shoal grass. Trends in percent cover could not be evaluated for unknown Halophila and drift algae due to insufficient data, and the model could not be fitted for manatee grass.

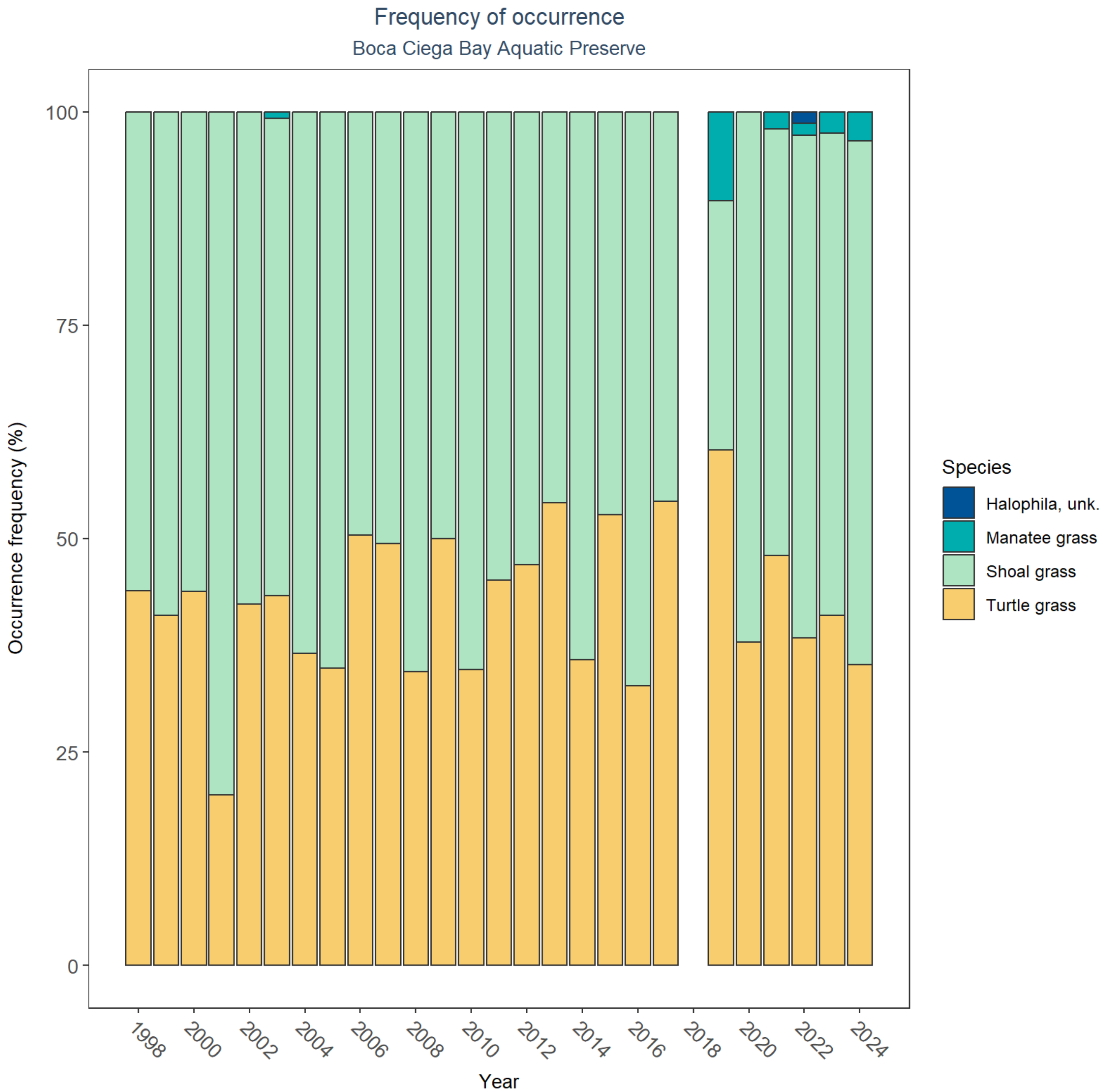

Figure 25: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Boca Ciega Bay Aquatic Preserve within the SAV_WC_Report:

- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity
- Secchi Depth

- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: DRAFT_SAV_WC_Report.pdf

Nekton

The data file used is: **All_NEKTON_Parameters-2026-Jun-04.txt**

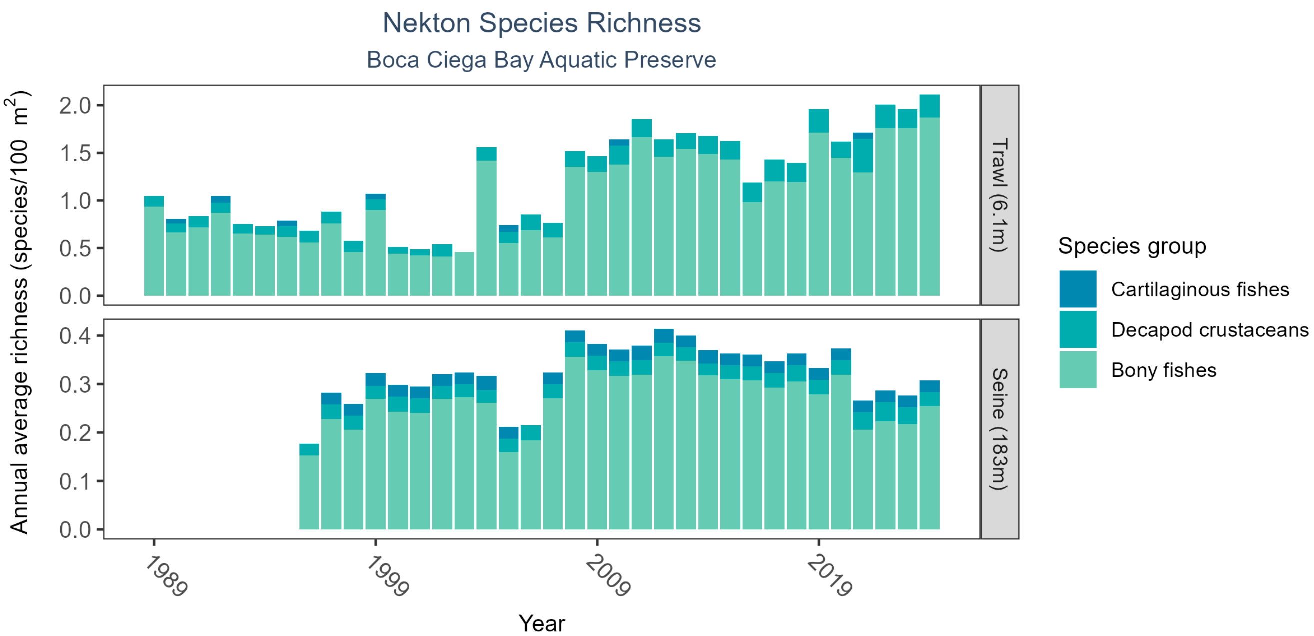

Figure 26: Bar graph(s) of annual average nekton richness over time for species groups occurring in at least 1% of samples. The bar colors represent species groups including bony fishes, cartilaginous fishes, decapod crustaceans (e.g., shrimps, crabs, and lobsters), and cephalopods (e.g., squid). Gear types and sizes are indicated in the panel label.

Table 28: Nekton Species Richness

Gear Type	*No. of Samples*	*No. of Years with Data*	*Period of Record*	*Median No. of Taxa*	*Mean No. of Taxa*
Trawl (6.1 m)	1830	36	1989 - 2024	0.45	0.77
Seine (183 m)	1655	29	1996 - 2024	0.15	0.19

The median annual number of taxa was 0.15 based on 1,655 observations collected by 183-meter seine between 1996 and 2024, and the median annual number of taxa was 0.45 based on 1,830 observations collected by 6.1-meter trawl between 1989 and 2024.

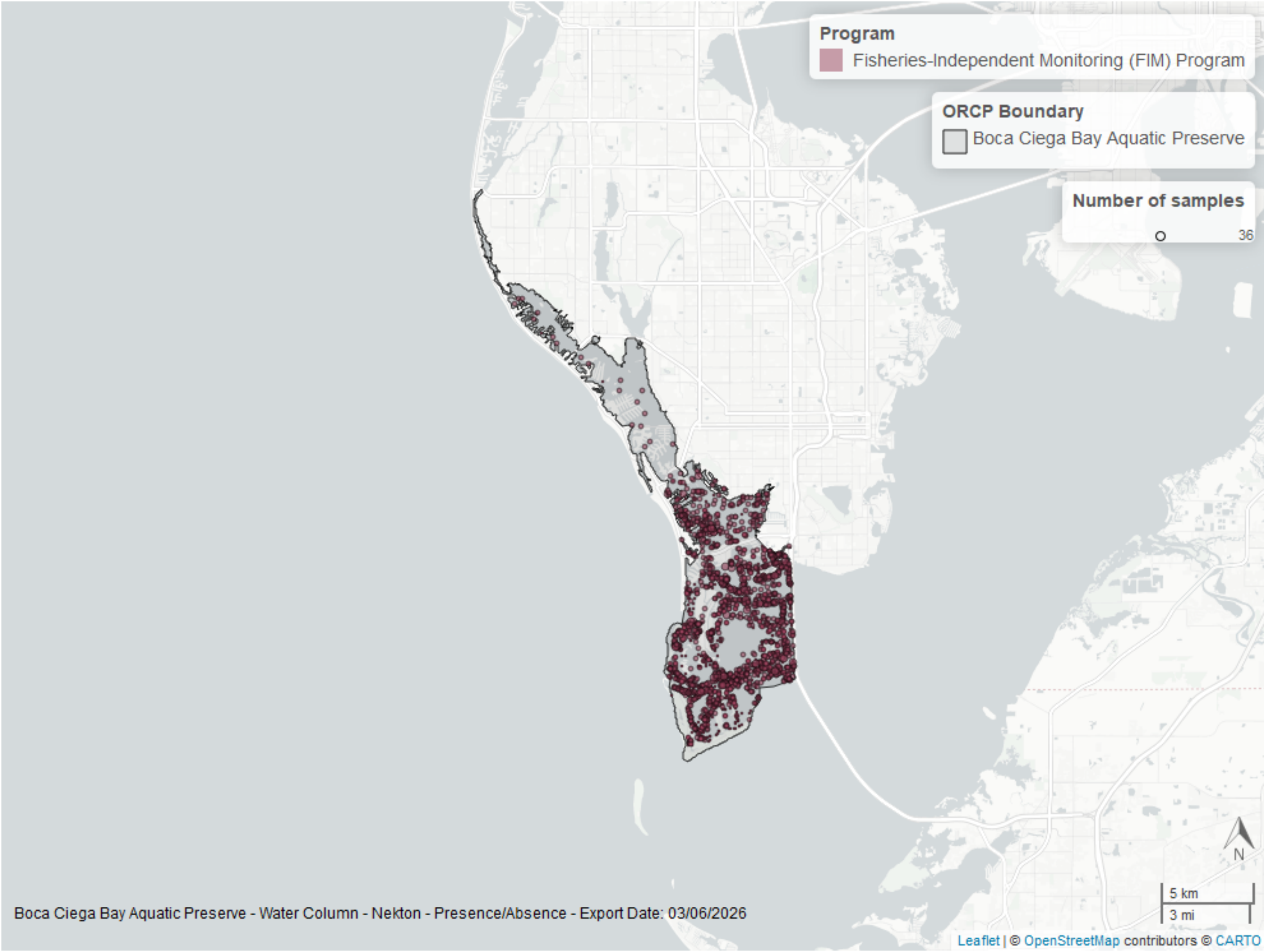

Figure 27: Map showing location of nekton sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Coastal Wetlands

The data file used is: **All__CW__Parameters-2026-Jun-04.txt**

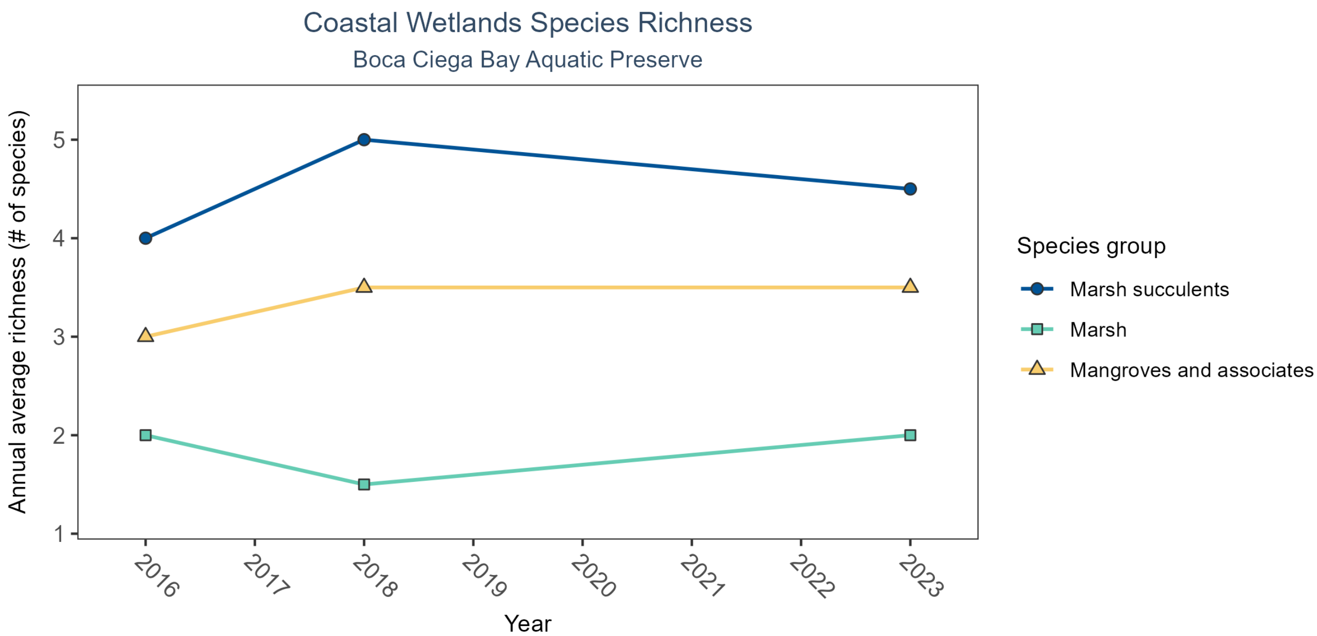

Figure 28: Line graph of annual average coastal wetlands species richness over time for mangroves and associates (triangles), marsh (squares), and marsh succulents (circles). If the time series by species group included more than one year of observations, a line connects data points for visualization.

Table 29: Coastal Wetlands Species Richness

Species Group	*No. of Samples*	*No. Years with Data*	*Period of Record*	*Median No. of Taxa*	*Mean No. of Taxa*
Mangroves and associates	6	3	2016 - 2023	3.5	3.33
Marsh	6	3	2016 - 2023	2.0	1.83
Marsh succulents	6	3	2016 - 2023	4.5	4.50

Between 2016 and 2023, the median annual number of species for mangroves and associates was 3.5 based on 6 observations. Between 2016 and 2023, the median annual number of species for marsh was 2 based on 6 observations. Between 2016 and 2023, the median annual number of species for marsh succulents was 4.5 based on 6 observations.

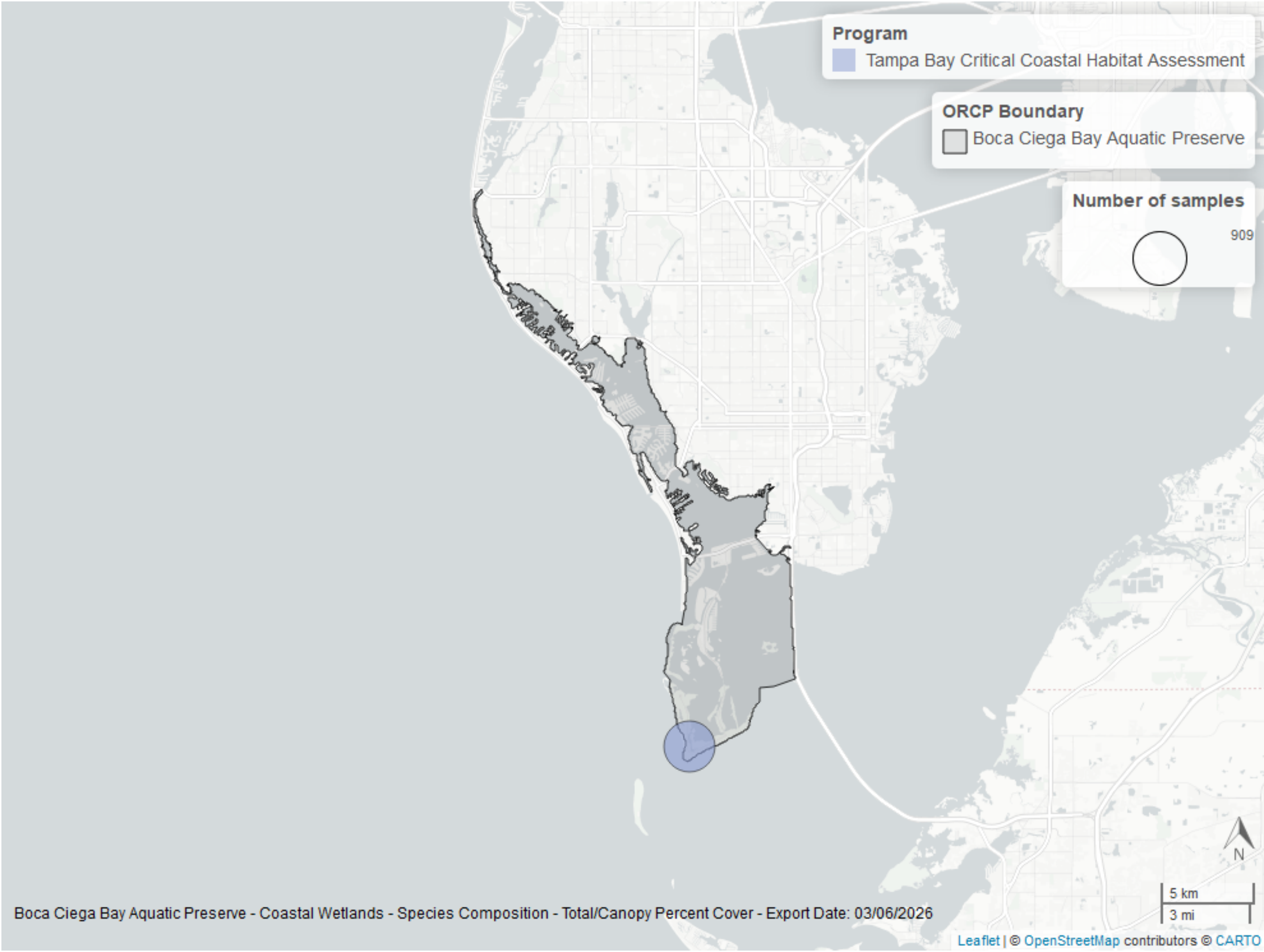

Figure 29: Map showing location of coastal wetlands sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Oyster

The data file used is: **All_OYSTER_Parameters-2026-Jun-04.txt**

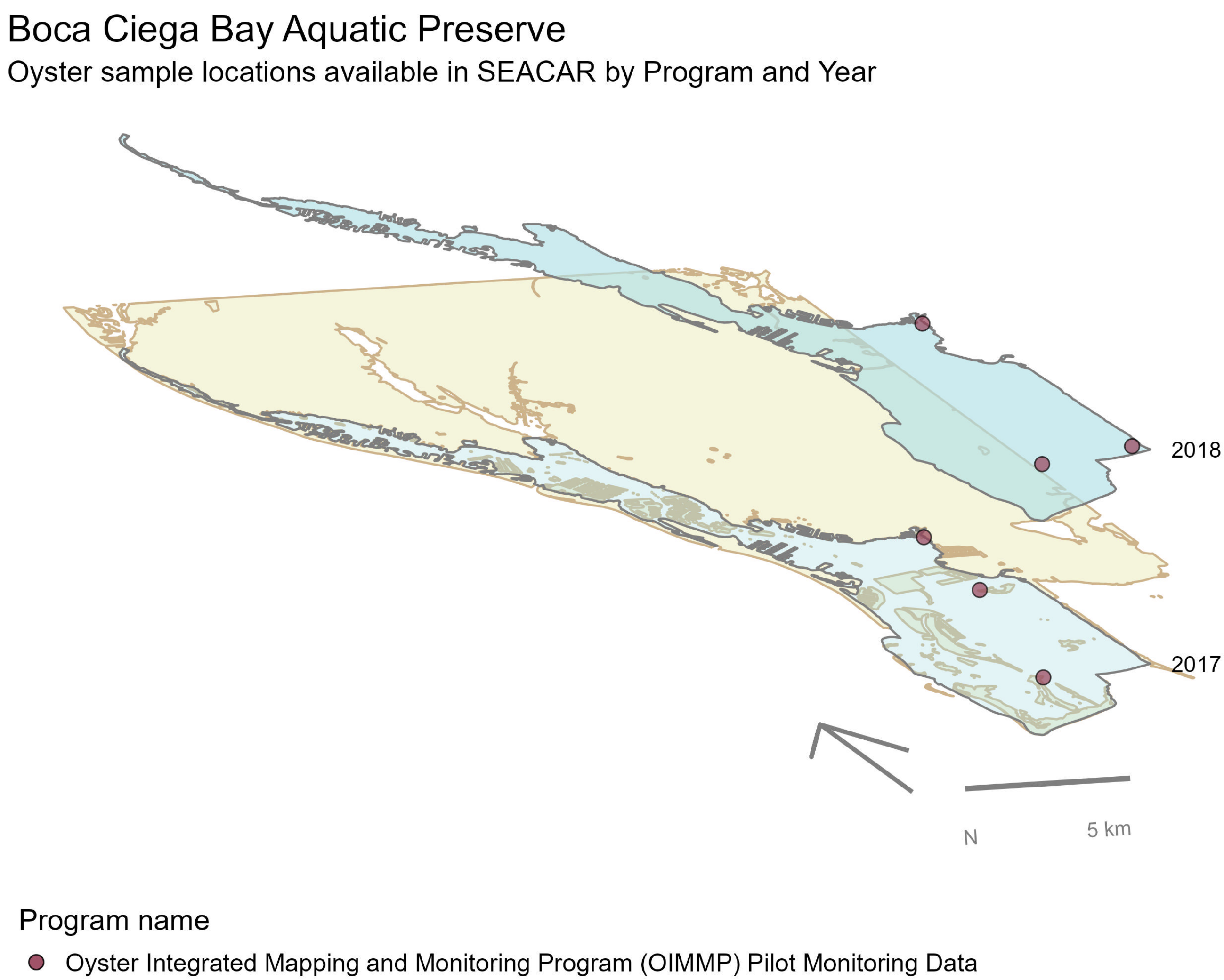

Figure 30: Maps showing the temporal scope of oyster sampling sites within the boundaries of *Boca Ciega Bay Aquatic Preserve* by program name.

Click here to view spatio-temporal plots on GitHub.

Density

For natural reefs, there was insufficient data to calculate a trend for density.

Natural

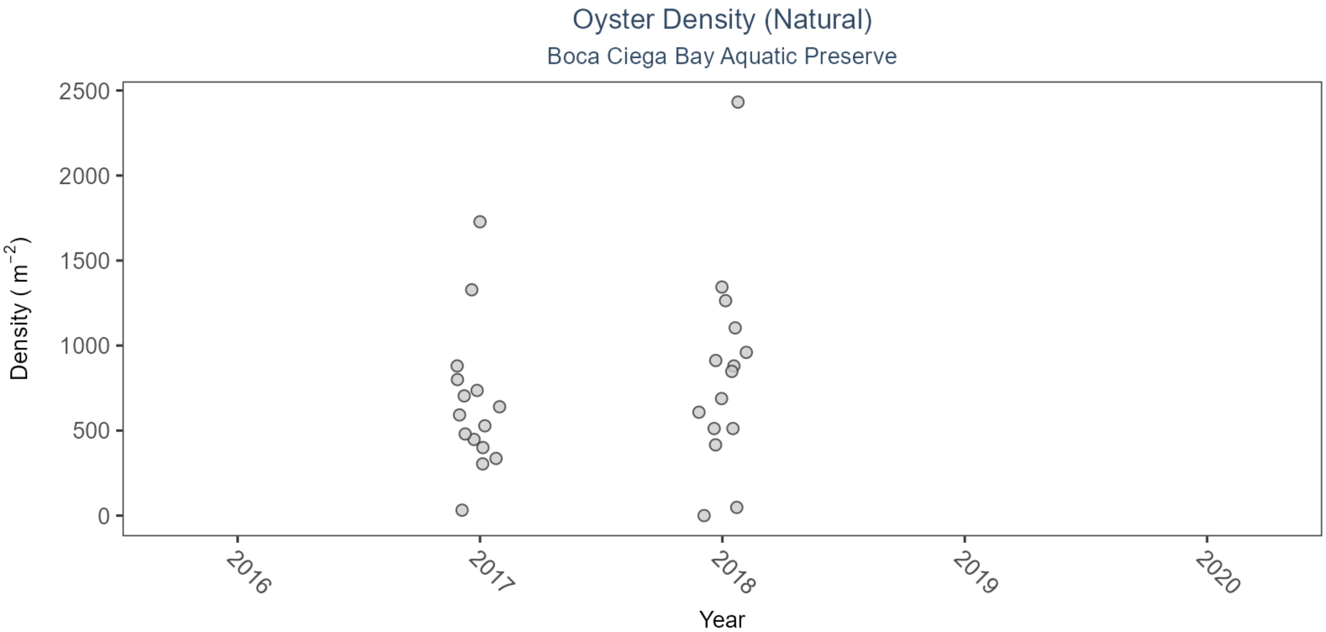

Figure 31: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 30: Model results for Oyster Density - Natural

Habitat Type	*Shell Type*	*Statistical Trend*	*Period of Record*	*Estimate*	*Standard Error*	*Credible Interval*
Natural	Live Oysters	Insufficient data	2017 - 2018	-	-	-

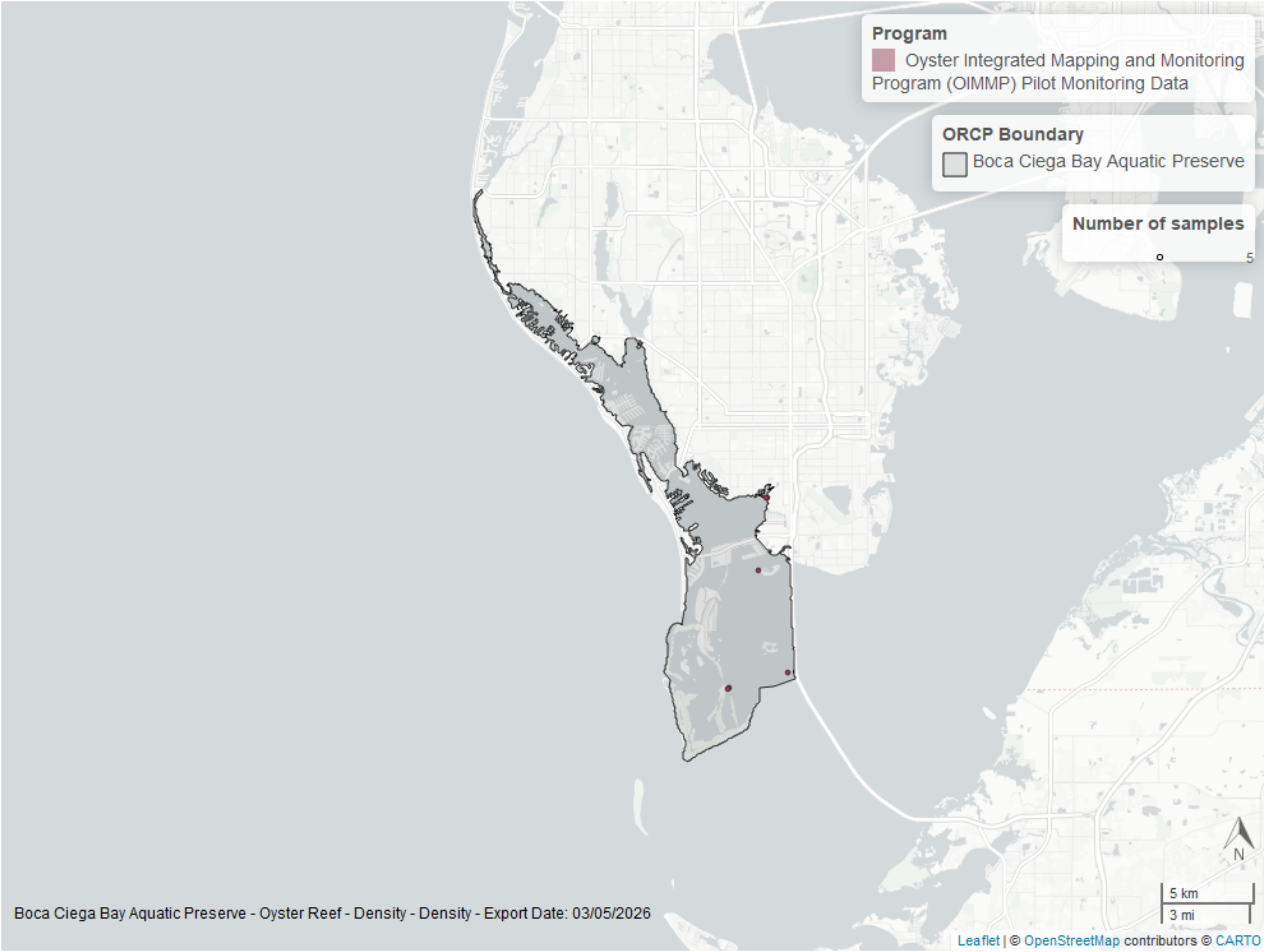

Figure 32: Map showing location of oyster density sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Percent Live

For natural reefs, there was insufficient data to calculate a trend for percent live cover.

Natural

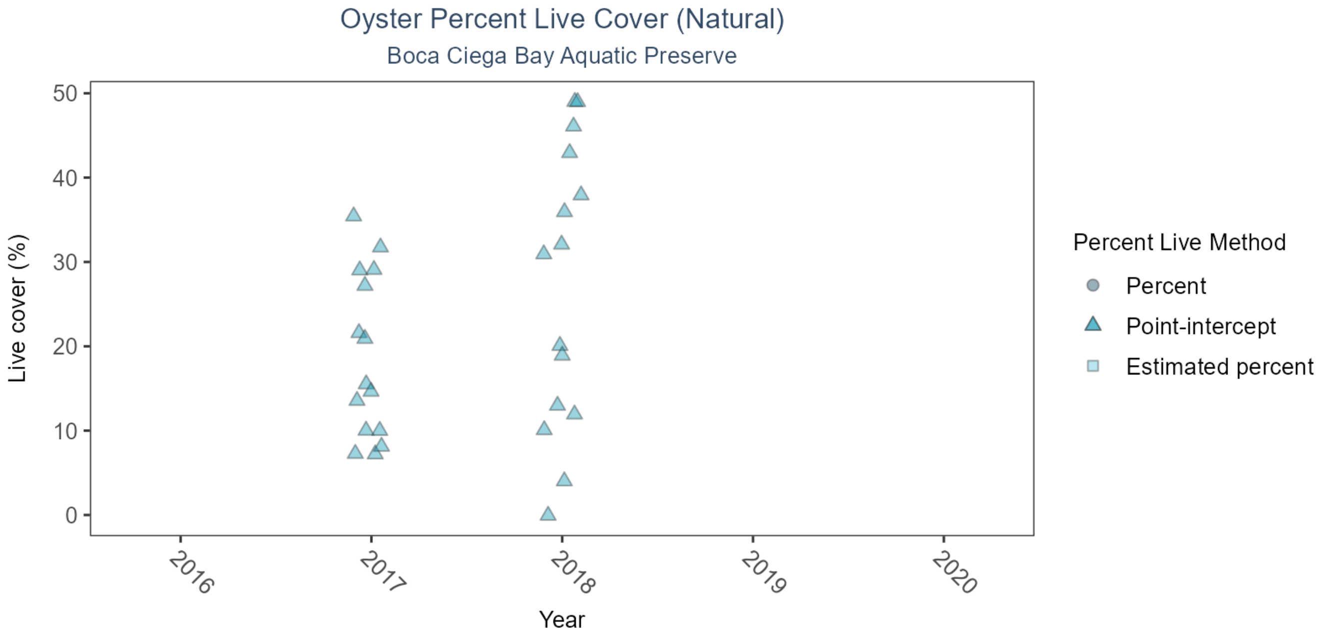

Figure 33: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 31: Model results for Oyster Percent Live - Natural

Habitat Type	*Shell Type*	*Statistical Trend*	*Period of Record*	*Estimate*	*Standard Error*	*Credible Interval*
Natural	Live Oysters	Insufficient data	2017 - 2018	-	-	-

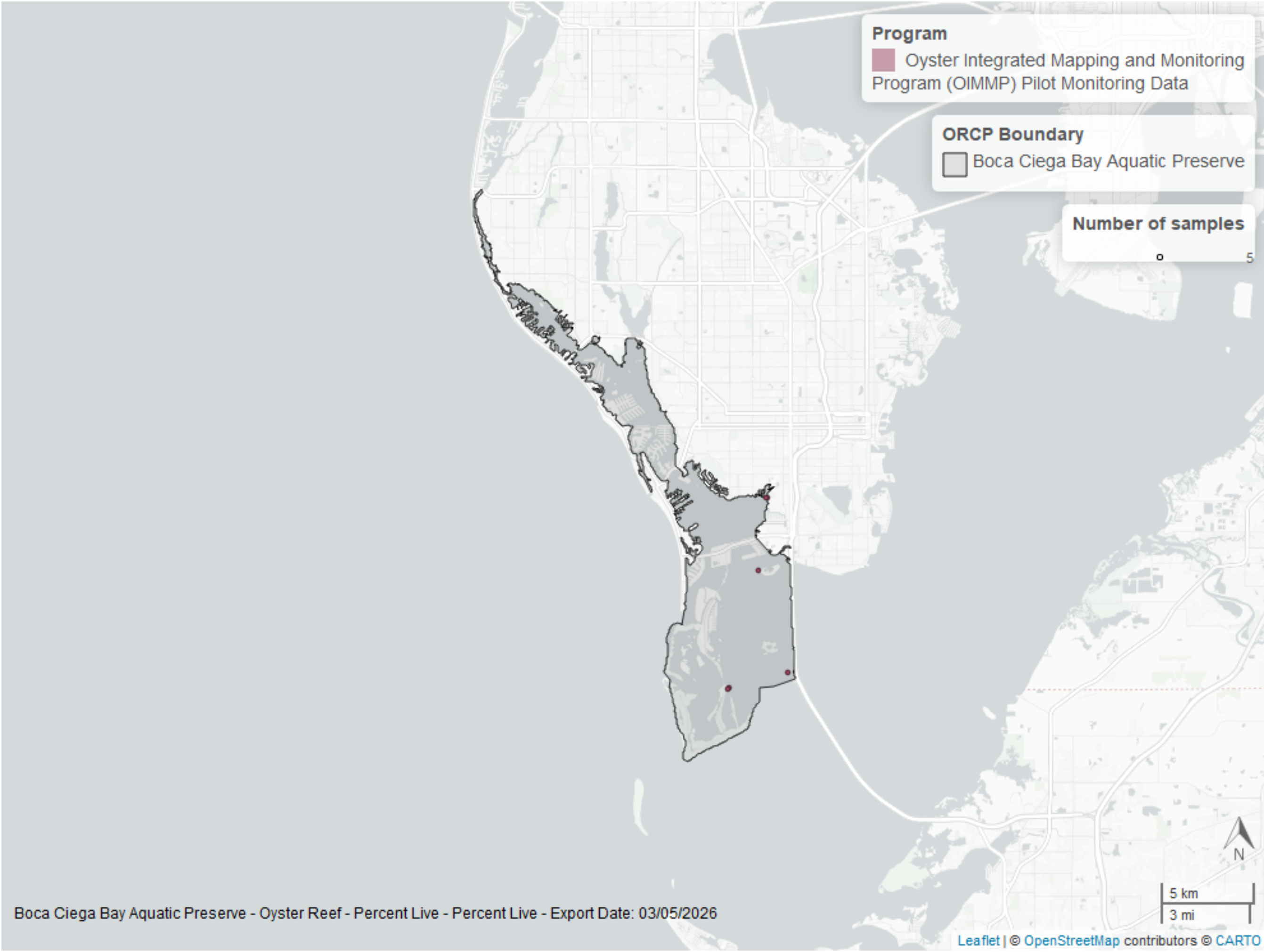

Figure 34: Map showing location of oyster percent live sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Shell Height

For natural reefs, there was insufficient data to calculate a trend for live oysters in either the 25-75mm or the >=75mm size class.

Natural

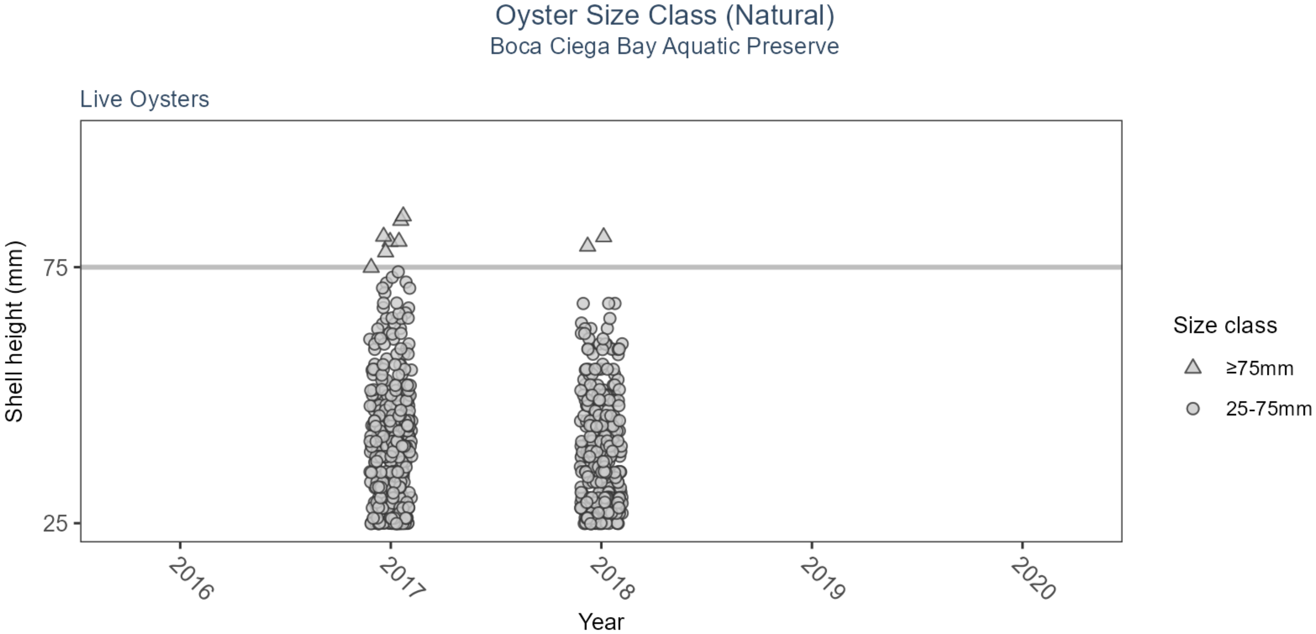

Table 32: Model results for Oyster Shell Height - Natural

Habitat Type	*Shell Type*	*SizeClass*	*Statistical Trend*	*Period of Record*	*No. of Samples*	*Estimate*	*Standard Error*	*Credible Interval*
Natural	Live Oysters	>75mm	Insufficient data	2017 - 2018	9	-	-	-
Natural	Live Oysters	25-75mm	Insufficient data	2017 - 2018	701	-	-	-

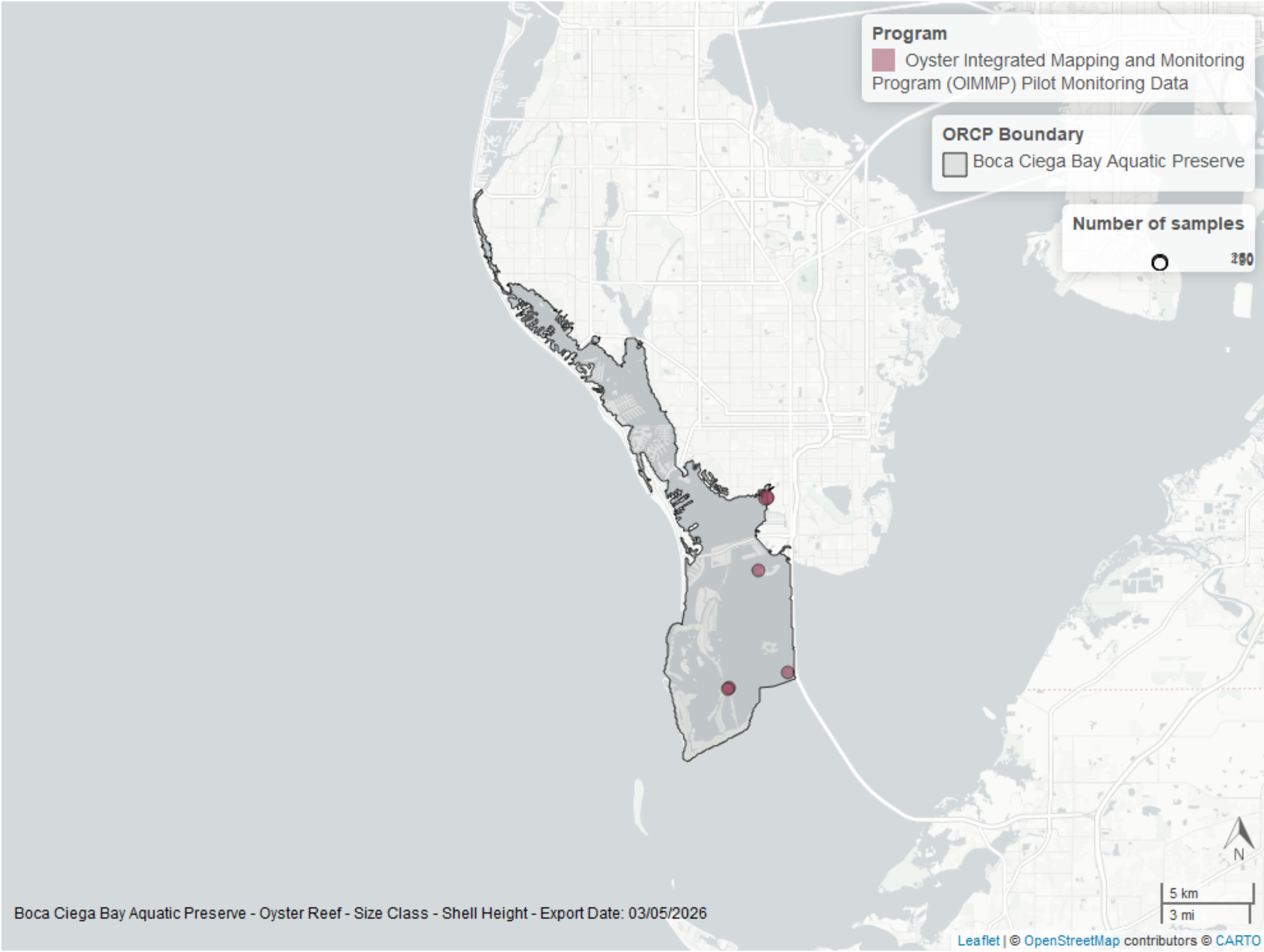

Figure 35: Map showing location of oyster shell height sampling locations within the boundaries of *Boca Ciega Bay Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Acanthophora sp.[1]
Acanthostracion quadricornis[3]
Achirus lineatus[3]
Aetobatus narinari[3]
Albula vulpes[3]
Alpheidae spp.[3]
Aluterus schoepfii[3]
Aluterus scriptus[3]
Anarchopterus criniger[3]
Anchoa cubana[3]
Anchoa hepsetus[3]
Anchoa lyolepis[3]
Anchoa mitchilli[3]
Anchoa spp.[3]
Ancylopsetta quadrocellata[3]
Anguilla rostrata[3]
Archosargus probatocephalus[3]
Argopecten irradians
Argopecten spp.
Ariopsis felis[3]
Astroscopus ygraecum[3]
Attached algae[1]
Avicennia germinans[2]
Bagre marinus[3]
Bairdiella chrysoura[3]
Bathygobius soporator[3]
Bothidae spp.[3]
Brevoortia spp.[3]
Calamus arctifrons[3]
Calamus penna[3]
Calamus proridens[3]
Calamus spp.[3]
Callinectes ornatus[3]
Callinectes sapidus[3]
Callinectes similis[3]
Callinectes spp.[3]
Caranx bartholomaei[3]
Caranx crysos[3]
Caranx hippos[3]
Caranx latus[3]
Caranx spp.[3]
Carcharhinus limbatus[3]
Caulerpa sertularioides[1]
Centropomus undecimalis[3]
Centropristis striata[3]
Chaetodipterus faber[3]
Chaetodon capistratus[3]
Chaetodon ocellatus[3]
Chasmodes saburrae[3]
Chelonia mydas[3]
Chilomycterus schoepfii[3]
Chloroscombrus chrysurus[3]
Citharichthys cornutus[3]
Citharichthys macrops[3]
Eucinostomus argenteus[3]
Eucinostomus gula[3]
Eucinostomus harengulus[3]
Eucinostomus spp.[3]
Eugerres plumieri[3]
Ficus aurea
Floridichthys carpio[3]
Fundulus grandis[3]
Fundulus similis[3]
Gerres cinereus[3]
Gobiesox strumosus[3]
Gobiidae spp.[3]
Gobionellus oceanicus[3]
Gobiosoma bosc[3]
Gobiosoma longipala[3]
Gobiosoma robustum[3]
Gobiosoma spp.[3]
Gracilaria sp.[1]
Haemulon aurolineatum[3]
Haemulon plumierii[3]
Haemulon spp.[3]
Halichoeres bivittatus[3]
Halodule wrightii[1]
Halophila sp.[1]
Harengula jaguana[3]
Hemicaranx amblyrhynchus[3]
Hemiramphus brasiliensis[3]
Hippocampus erectus[3]
Hippocampus zosterae[3]
Hypleurochilus caudovittatus[3]
Hypnea[1]
Hyporhamphus meeki[3]
Hyporhamphus spp.[3]
Hyporhamphus unifasciatus[3]
Hypsoblennius hentz[3]
Kyphosus sectatrix[3]
Kyphosus spp.[3]
Lachnolaimus maximus[3]
Lactophrys trigonus[3]
Lagodon rhomboides[3]
Laguncularia racemosa[2]
Leiostomus xanthurus[3]
Limulus polyphemus
Lobotes surinamensis[3]
Lucania parva[3]
Lutjanus analis[3]
Lutjanus apodus[3]
Lutjanus griseus[3]
Lutjanus spp.[3]
Lutjanus synagris[3]
Lycium carolinianum
Lyngbya sp.
Maytenus phyllanthoides
Menidia spp.[3]
No fish
No grass in quadrat[1]
Ocyurus chrysurus[3]
Ogcocephalus cubifrons[3]
Ogcocephalus spp.[3]
Oligoplites saurus[3]
Ophidion grayi[3]
Ophidion holbrookii[3]
Ophidion josephi[3]
Opisthonema oglinum[3]
Opsanus beta[3]
Oreochromis aureus[3]
Orthopristis chrysoptera[3]
Ostraciidae spp.[3]
Paraclinus fasciatus[3]
Paraclinus marmoratus[3]
Paralichthys albigutta[3]
Penaeidae spp.[3]
Penaeus duorarum[3]
Peprilus burti[3]
Peprilus paru[3]
Phoenix reclinata
Poecilia latipinna[3]
Pogonias cromis[3]
Pomatomus saltatrix[3]
Portunus spp.[3]
Prionotus scitulus[3]
Prionotus spp.[3]
Prionotus tribulus[3]
Pseudocrenilabrinae[3]
Rachycentron canadum[3]
Rhinoptera bonasus[3]
Rhizophora mangle[2]
Rimapenaeus constrictus[3]
Sabal palmetto
Sardinella aurita[3]
Sarotherodon melanotheron[3]
Schinus terebinthifolia
Sciaenops ocellatus[3]
Scomberomorus maculatus[3]
Scorpaena brasiliensis[3]
Selene vomer[3]
Serraniculus pumilio[3]
Serranus subligarius[3]
Sicyonia brevirostris[3]
Sicyonia laevigata[3]
Sicyonia spp.[3]
Sicyonia typica[3]
Sphoeroides nephelus[3]
Sphoeroides spengleri[3]
Sphyraena barracuda[3]
Sphyraena borealis[3]
Sphyraena guachancho[3]
Sphyrna tiburo[3]

Conocarpus erectus[2]	Menippe mercenaria[3]	Stephanolepis setifer[3]
Cynoscion arenarius[3]	Menippe spp.[3]	Strongylura marina[3]
Cynoscion nebulosus[3]	Menticirrhus americanus[3]	Strongylura notata[3]
Cyprinodon variegatus[3]	Menticirrhus littoralis[3]	Strongylura timucu[3]
Diapterus auratus[3]	Menticirrhus saxatilis[3]	Syacium papillosum[3]
Diodon holocanthus[3]	Menticirrhus spp.[3]	Symphurus plagiusa[3]
Diplectrum formosum[3]	Microgobius gulosus[3]	Syngnathus floridae[3]
Diplectrum spp.[3]	Microgobius spp.[3]	Syngnathus louisianae[3]
Diplodus holbrookii[3]	Microgobius thalassinus[3]	Syngnathus scovelli[3]
Dorosoma petenense[3]	Micropogonias undulatus[3]	Syngnathus springeri[3]
Drift algae[1]	Monacanthus ciliatus[3]	Synodus foetens[3]
Drift red algae[1]	Mugil cephalus[3]	Syringodium filiforme[1]
Echeneis naucrates[3]	Mugil curema[3]	Thalassia testudinum[1]
Echeneis neucratoides[3]	Mugil spp.[3]	Trachinotus carolinus[3]
Echeneis spp.[3]	Mugil trichodon[3]	Trachinotus falcatus[3]
Elops saurus[3]	Mycteroperca bonaci[3]	Trinectes maculatus[3]
Engraulidae spp.[3]	Mycteroperca microlepis[3]	Tylosurus crocodilus[3]
Epinephelus itajara[3]	Myrophis punctatus[3]	Unidentified species
Epinephelus morio[3]	Narcine bancroftii[3]	Urophycis floridana[3]
Etropus crossotus[3]	Nicholsina usta[3]	Acanthophora sp.[1]

1 - Submerged Aquatic Vegetation, 2 - Coastal Wetlands, 3 - Nekton

References

1. Florida Department of Environmental Protection (DEP). Florida STORET / WIN. (2024).
2. U.S. Environmental Protection Agency (EPA). EPA STOrage and RETrieval Data Warehouse (STORET)/WQX. (2023).
3. U.S. Environmental Protection Agency (EPA); Office of Research and Development. Environmental Monitoring Assessment Program. (2004).
4. U.S. Environmental Protection Agency (EPA); Office of Water; National Oceanic and Atmospheric Administration (NOAA); U.S. Geological Survey (USGS); U.S. Fish and Wildlife Service (USFWS); National Estuary Program (NEP); coastal states. National Aquatic Resource Surveys, National Coastal Condition Assessment. (2021).
5. Florida Fish and Wildlife Conservation Commission (FWC). Fisheries-Independent Monitoring (FIM) Program. (2022).
6. Tampa Bay Estuary Program. Tampa Bay Benthic Monitoring. (2022).
7. Florida Fish and Wildlife Conservation Commission (FWC); Florida Fish and Wildlife Research Institute (FWRI). Harmful Algal Bloom Marine Observation Network. (2018).
8. National Oceanic and Atmospheric Administration (NOAA); Center for Coastal Monitoring and Assessment. National Status and Trends Mussel Watch. (2000).
9. Pinellas County (Florida); Department of Environmental Management; Tampa Bay Estuary Program (TBEP). Western Pinellas County Seagrass Monitoring. (2019).
10. Tampa Bay Interagency Seagrass Monitoring Program; Tampa Bay Estuary Program (TBEP). Tampa Bay Seagrass Monitoring. (2023).